# Supporting Information for "Retaining incomplete river-temperature records: time support of finite-record heat slopes"

Author information to be completed by the authors

**Contents of this file**

1. Text S1–S39.

2. Tables S1–S71, including external-reference Table S46.

The following sections keep the denominators, qualification ledger and negative boundary results needed by the main claims. Numerical-budget, external-recruitment and release-rule explorations are compressed to that supporting role; they are not additional primary results. Source data, model outputs and software require the separate access arrangements described in the article's Open Research section; this file does not substitute for those materials.

## S1. Design and information separation

The source comparison uses twenty repetitions of masks and group-role assignment on the same assembled archive; these are not independent climates. Held test connected groups are excluded from estimator fitting, interval calibration, risk learning and rule calibration. In the original partitioned forward models, other training-only groups in a

named target region can remain in source roles. The source-role audit and subsequent strict-region sensitivities distinguish this exception from the held target-group exclusion.

In each whole-archive setting, selected gaps in the target station are masked together before local daily fitting. Unmasked target observations and the designated donor observations remain available. The source annual model is fitted separately and held fixed within that archive reanalysis. The first seven years are observed burn-in; the forward evaluation masks only years from 2010 onward. Twenty repetitions at the same station are not twenty independent rivers. Initial seven-station stress pilots are followed by a 41-station forward sensitivity; the heat-associated extension fixes the number of missing years but can change their dates and the number of missing days.

**Correction to historical design metadata**

The 3,320 saved whole-archive `design.json` files are historical records and were not rewritten. Their field `annual_model` states that "held region and connected groups excluded." That statement is correct for the held test connected groups, but it is not an unconditional region exclusion. The original forward HUC17 estimator retained one other same-region training group, and NVE interval calibration retained two other same-region groups. The paper therefore distinguishes the held-test-group main analysis from the strict-region sensitivity. This correction applies to the historical metadata; it does not change any saved prediction. A correction manifest is available in `results/wrr_adversarial_review_20261027/metadata_correction/`.

## S2. Observation-rule accounting

The rule table uses every eligible station-year, not only the selected witnesses. A reported window is not a complete-year H30, and an accepted incomplete year can contain an unobserved higher peak. The "new facts" column therefore compares reported windows with the maximum reported window among the other accepted years; it does not prove a full-record ordering. The measurement-margin sensitivity treats a window ordering as unresolved when the reported difference is at most twice the assumed absolute error on each window. These margins are assumptions, not sensor-error estimates.

## S3. Joint-rank numerical convergence

This is a fixed-marginal numerical-budget check, not the source-role experiment. It keeps the saved annual quantiles, observed bounds and evaluation years and increases only the stratified sample count through 256, 1,024 and 4,096 on the difficult forward controlled gaps. Three random seeds index the rank construction. The rank sequence is generated at 4,096 draws and truncated for smaller counts; annual marginals are regenerated separately for each sample size and remain specific to each estimator. Coverage, width and interval score are reported with the across-seed range. The range is not a confidence interval, and the check uses the same opened data as the main analysis. A 0.3-point coverage rule and 1% width and interval-score rules are applied to the three-seed mean. Five of six method–construction pairs meet the coverage rule on that mean. The largest three-seed-mean coverage change is 0.36 percentage points, and the largest single-seed change is 0.815 percentage points. Averaging seeds can cancel opposing changes, so these rules are not a resolution guarantee and the constructions are not ranked. At 4,096 samples, nominal

95% trend coverage remains 87.4–89.9%. The later source-budget experiment instead regenerates annual residual-bridge distributions and re-estimates source-role offsets.

## S4. Scoring and interval construction

For ensemble $x_1, \ldots, x_M$ and outcome $y$, the continuous ranked probability score is

$$M^{-1} \sum_{m} |x_m - y| - (2M^2)^{-1} \sum_{m,l} |x_m - x_l|.$$

Threshold-weighted scoring replaces samples and outcome by their maxima with the training H30 threshold. The 90% interval score adds twenty times the distance outside the interval to its width. A 95% trend interval score uses the corresponding factor forty. Missing predictions remain unavailable rather than being assigned zero error.

Both annual estimators use the observed-window support projection where a finite bound exists. The projection is applied to the annual distribution and then carried into the slope; a trend interval is not projected onto an annual lower bound. Seven stored quantiles do not determine their outer tails; the normal-spacing continuation and the alternative constant tail are assumptions. Within one estimator, cross-year rank constructions keep that estimator's annual marginal samples fixed. The two estimators have different annual marginals. The constructions are not a posterior for the fitted annual model. The source-only corrections and the post-hoc propagation reported below do not establish a finite-sample or independent-climate coverage guarantee.

In whole-archive annual coverage and the peak-visibility analysis, inclusive endpoint comparisons use an absolute $10^{-8}\,^{\circ}$C numerical tolerance. After detecting boundary sensitivity in the USGS exploration, we compared this convention with exact floating-point comparisons and a $10^{-10}\,^{\circ}$C tolerance on the same saved two-estimator common cases. For

the forward empirical-calendar subset, D2 covers 3,610 of 3,674 repeated masked cases with either tolerance, but an exact comparison covers 2,037; all 1,573 changed comparisons lie in the reference-at-observed-bound stratum. Across all four family–mode combinations and both estimators, the two nonzero tolerances agree on every common-case classification. An independent scalar audit found the largest strict-comparison endpoint gap among changed method–case comparisons was $8.18 \times 10^{-14}\,^{\circ}\mathrm{C}$. These tolerances concern arithmetic on the reported target, not physical measurement uncertainty, and masks from a station-year are not independent. The pooled common-case comparison differs in weighting and availability from the equal-archive means used elsewhere; no original saved coverage was replaced.

The historical single-year source comparison also uses an absolute $10^{-8}\,^{\circ}\mathrm{C}$ endpoint tolerance after support projection. A post-hoc check reproduced the published coverage for all 20 saved source repetitions, two methods, two gap families and both overlap strata. In the controlled-gap, high-overlap stratum, $10^{-10}$ and $10^{-8}\,^{\circ}\mathrm{C}$ give identical coverage classifications and median D2 and stochastic coverage remains 89.3% and 89.8%, respectively. Outside the high-overlap stratum there are 14 repeated method–case comparisons where the two tolerances differ, with endpoint gaps between $1.89 \times 10^{-9}$ and $9.93 \times 10^{-9}\,^{\circ}\mathrm{C}$. An independent scalar implementation reproduced the 160 method–family–stratum–seed counts. These are the same historical source predictions, not independent sensor validation or a population calibration guarantee.

For the exploratory release-then-trend analysis, an observed lower bound $L$ contributes $\log \Phi\{(\mu - L)/\sigma\}$ to a Gaussian working likelihood. Released numerical estimates are treated as exact, unknown cases remain unavailable, and the working interval includes regression variability. It is not a calibrated missingness-only interval. Tables S2 and S3 retain

the negative matched-retention results; they are not a proposed operational publication system.

## S5. Core source comparisons

### S5.1. Additional observation-only corroboration

We obtained approved daily-mean USGS temperature records at ten additional metadata-selected stream stations. None was a source-model target. Seven candidates lie in HUC2 regions 13 or 20, absent from the source station regions; three are additional station checks within already represented HUC2 regions. The candidates and all qualification failures are retained. This unregistered intake was not a random sample or a blinded confirmation, and the reference histories are shorter and sparser than the original long-record panel.

The unchanged four-complete-year reconstruction criterion yields zero eligible numerical-recovery references. The observation-only claim needs a different comparison: an incomplete year can contradict the maximum of the complete-year record whenever at least one approved-complete reference year exists. We retain every approved finite reference value, including reference years that an extra heuristic-QC filter might remove. Candidate windows must contain thirty consecutive approved, nonestimated daily values after source QC, and the incomplete year must have an actual absent, null, contradictory or unapproved provider day. Heuristic censoring alone cannot create a witness.

There are four stations with a nonempty approved-complete reference, each with only one to three reference years. At 3 of these stations, 9 incomplete station-years contain verified windows exceeding every approved-complete H30 in the analyzed record (Figure 2 of the main article). For example, the Rio Mora 2020 window exceeds the complete-only record by approximately 0.883°C, and a Pecos River 2020 window by 0.643°C. These facts

**Table S1.** Dangerous source-case summaries: the gap overlaps at least half the training-period peak window. Entries are medians across 20 repetitions sharing the source archive. Method-specific availability can differ; paired case comparisons are reported separately. Coverage and CRPS are not defined for point-only methods.

| Family | Method | Availability | MAE | 90% coverage | CRPS |
|---|---|---|---|---|---|
| calendar | D2 | 100.0% | 0.451 | 86.7% | 0.329 |
| calendar | functional | 100.0% | 0.476 | – | – |
| calendar | plug-in | 100.0% | 0.564 | – | – |
| calendar | classical | 99.9% | 0.463 | 86.0% | 0.336 |
| crossed | D2 | 100.0% | 0.337 | 89.3% | 0.246 |
| crossed | functional | 100.0% | 0.355 | – | – |
| crossed | plug-in | 100.0% | 0.428 | – | – |
| crossed | classical | 99.7% | 0.336 | 89.8% | 0.245 |

**Table S2.** Held-region source-calibrated comparison of the three-output and score-only rules. The score cutoff is chosen from disjoint source rule-calibration groups. Retention differs in the evaluated target cases; errors are conditional on numerical publication.

| Family | Method | Eligible | Published | Retention | Error rate |
|---|---|---|---|---|---|
| calendar | score only | 31,672 | 26,896 | 84.9% | 0.0% |
| calendar | three outputs | 31,672 | 27,090 | 85.5% | 0.7% |
| crossed | score only | 114,012 | 80,980 | 71.0% | 3.9% |
| crossed | three outputs | 114,012 | 82,169 | 72.1% | 4.9% |

**Table S3.** Retrospective equal numerical-count comparison. The score threshold uses the evaluated feature distribution and the three-output count, so this is transductive. Equal numerical counts do not equate retained bounds or unknown states.

| Condition | Family | Eligible | Published by each | Three failures | Score failures | Three / score error |
|---|---|---|---|---|---|---|
| air substitution | calendar | 31,672 | 88 | 19 | 23 | 21.6% / 26.1% |
| clean | calendar | 31,672 | 27,090 | 190 | 4 | 0.7% / 0.0% |
| clean | crossed | 114,012 | 82,169 | 4026 | 3249 | 4.9% / 4.0% |
| future | crossed | 27,846 | 17,583 | 1924 | 1220 | 10.9% / 6.9% |

**Table S4.** Annual and trend coverage on the common archive cohort. Annual coverage weights cases by incomplete years. Tail and rank experiments use different random draws. Within one estimator the three rank constructions share that estimator's annual marginals; the two estimators do not share one annual distribution. Repetitions share the same source archive.

| Family | Mode | Method | Cases | Groups | Annual 90% | Constant 95% | Normal 95% | Independent 95% | Shared 95% | Blocks 95% |
|---|---|---|---|---|---|---|---|---|---|---|
| calendar | forward | D2 | 818 | 17 | 98.2% | 94.6% | 95.7% | 96.0% | 96.1% | 96.3% |
| calendar | forward | classical | 699 | 17 | 98.2% | 94.6% | 95.3% | 95.4% | 95.4% | 95.7% |
| calendar | offline | D2 | 840 | 18 | 98.5% | 92.3% | 94.9% | 95.0% | 95.1% | 94.9% |
| calendar | offline | classical | 835 | 18 | 97.4% | 91.7% | 93.3% | 93.2% | 93.3% | 93.3% |
| crossed | forward | D2 | 818 | 17 | 90.1% | 83.0% | 88.8% | 88.0% | 87.9% | 89.0% |
| crossed | forward | classical | 736 | 17 | 88.1% | 81.4% | 86.3% | 86.3% | 86.1% | 87.2% |
| crossed | offline | D2 | 840 | 18 | 94.1% | 91.7% | 94.2% | 94.0% | 94.5% | 94.8% |
| crossed | offline | classical | 838 | 18 | 92.6% | 86.4% | 90.7% | 90.2% | 91.6% | 91.2% |

**Table S5.** Evidence map. Each claim is paired with the comparison and metric used here and with the boundary that limits it. The table indexes the results; it is not an additional test.

| Claim | Primary comparison | Metric | Boundary |
|---|---|---|---|
| Observed windows can overturn a complete-only record | Complete-year H30 vs. fully observed 30-day window $L$ | Count of $L$ above the complete-year maximum | Reported measurements only; not the unique warmest year |
| Target recovery beats a daily plug-in | D2 vs. plug-in, functional debiasing and calibrated stochastic regression | H30 MAE, CRPS and 90% coverage | No unique gain over calibrated stochastic regression |
| Retaining years improves finite-record trends | Warm-season rule vs. all supported years at the same estimator | Trend RMSE; arithmetic four-cell retention/estimator contrasts | Contrasts are setting-specific, not causal year-count shares |
| Annual and trend uncertainties need distinct evaluation | Bounded normal-tail vs. constant-tail marginals; three rank couplings | Coverage, width, interval score and availability | Tail, annual and dependence errors are not uniquely separated |
| Numerical release has a risk–retention trade-off | Fixed score cap vs. three-output rule | Selected risk, retention and group excess | A raw individual score is not a group-risk guarantee |
| Visible diagnostics do not restore extrapolation safety | Clean vs. hot-year, future-year and air-exposure stress | Selected risk, coverage and retention | Synthetic stress severity; not an archive prevalence |

**Table S6.** Every eligible station-year under the compared completeness rules. "Number" is a complete-year H30; "bound" is an incomplete year with a fully observed 30-day window; "unknown" has neither. "New facts" counts admitted incomplete years whose observed window exceeds the maximum reported window among other accepted years.

| Rule | Eligible | Admitted | Number | Bound | Unknown | New facts | Excluded facts |
|---|---|---|---|---|---|---|---|
| Complete year (100%) | 2,700 | 2,139 | 2,139 | 0 | 0 | 0 | 35 |
| Annual coverage $\geq$95% | 2,700 | 2,591 | 2,139 | 452 | 0 | 18 | 3 |
| Station warm-season $\geq$90% | 2,700 | 2,634 | 2,139 | 495 | 0 | 19 | 2 |
| Fixed warm-season $\geq$90% | 2,700 | 2,635 | 2,139 | 496 | 0 | 19 | 2 |
| July–August complete | 2,700 | 2,509 | 2,139 | 370 | 0 | 18 | 3 |
| Longest gap $\leq$49 d (context) | 2,700 | 2,653 | 2,139 | 514 | 0 | 19 | 1 |

A reported window is not a complete-year H30. A rule-accepted incomplete year can contain a higher unobserved peak, so the last two columns compare reported windows only. The 49-day row is a contextual Johnson comparison, not a validated H30 rule.

**Table S7.** Measurement-margin sensitivity for the station-specific warm-season rule. A reported window ordering is counted as resolved only when the observed difference exceeds twice the assumed absolute error on each window. The margins are assumptions, not sensor-error estimates.

| Assumed absolute error (°C) | Status | Facts | Resolved above $2\epsilon$ | Not resolved |
|---|---|---|---|---|
| 0.00 | admitted | 19 | 19 | 0 |
| 0.00 | excluded | 2 | 2 | 0 |
| 0.01 | admitted | 19 | 18 | 1 |
| 0.01 | excluded | 2 | 2 | 0 |
| 0.05 | admitted | 19 | 15 | 4 |
| 0.05 | excluded | 2 | 2 | 0 |
| 0.10 | admitted | 19 | 8 | 11 |
| 0.10 | excluded | 2 | 0 | 2 |
| 0.20 | admitted | 19 | 4 | 15 |
| 0.20 | excluded | 2 | 0 | 2 |
| 0.50 | admitted | 19 | 1 | 18 |
| 0.50 | excluded | 2 | 0 | 2 |

corroborate record omission without fitting any imputer. They do not estimate an omission incidence, verify a unique warmest year, establish external reconstruction accuracy, or validate trend or risk guarantees. Every witness includes the thirty original approved values; agency reporting and measurement validity remain assumptions.

**Table S8.** Numerical convergence of the three fixed-marginal joint-rank constructions on the difficult forward controlled gaps. Annual quantiles, bounds and evaluation years are fixed. The range is across three random seeds; it is not a confidence interval.

| Method | Construction | Draws | Coverage (%) | Width (°C/dec) | Interval score |
|---|---|---|---|---|---|
| D2 | independent | 256 | 88.3 [88.0, 88.5] | 0.450 [0.449, 0.451] | 0.855 [0.849, 0.860] |
| D2 | independent | 1024 | 89.2 [89.0, 89.4] | 0.457 [0.457, 0.458] | 0.845 [0.842, 0.847] |
| D2 | independent | 4096 | 89.2 [89.1, 89.2] | 0.459 [0.459, 0.459] | 0.842 [0.839, 0.844] |
| D2 | residual blocks | 256 | 89.2 [88.6, 89.6] | 0.451 [0.450, 0.451] | 0.840 [0.828, 0.851] |
| D2 | residual blocks | 1024 | 89.8 [89.7, 89.9] | 0.456 [0.456, 0.456] | 0.828 [0.825, 0.830] |
| D2 | residual blocks | 4096 | 89.7 [89.6, 89.9] | 0.457 [0.457, 0.458] | 0.825 [0.825, 0.826] |
| D2 | shared parameters | 256 | 88.4 [88.3, 88.6] | 0.444 [0.444, 0.445] | 0.855 [0.845, 0.863] |
| D2 | shared parameters | 1024 | 88.7 [88.3, 88.9] | 0.450 [0.449, 0.451] | 0.840 [0.835, 0.846] |
| D2 | shared parameters | 4096 | 88.7 [88.5, 88.9] | 0.452 [0.451, 0.452] | 0.835 [0.834, 0.835] |
| Classical | independent | 256 | 86.3 [86.0, 86.7] | 0.384 [0.383, 0.384] | 0.856 [0.849, 0.866] |
| Classical | independent | 1024 | 87.5 [87.2, 87.6] | 0.390 [0.390, 0.390] | 0.853 [0.852, 0.853] |
| Classical | independent | 4096 | 87.6 [87.5, 87.6] | 0.391 [0.391, 0.391] | 0.851 [0.851, 0.852] |
| Classical | residual blocks | 256 | 86.9 [86.5, 87.1] | 0.384 [0.383, 0.385] | 0.840 [0.837, 0.845] |
| Classical | residual blocks | 1024 | 87.6 [87.4, 87.9] | 0.389 [0.389, 0.390] | 0.837 [0.835, 0.839] |
| Classical | residual blocks | 4096 | 88.0 [87.8, 88.2] | 0.390 [0.390, 0.391] | 0.837 [0.836, 0.839] |
| Classical | shared parameters | 256 | 86.3 [85.9, 86.5] | 0.380 [0.380, 0.381] | 0.867 [0.864, 0.868] |
| Classical | shared parameters | 1024 | 87.3 [87.0, 87.5] | 0.385 [0.385, 0.385] | 0.856 [0.854, 0.859] |
| Classical | shared parameters | 4096 | 87.5 [87.4, 87.6] | 0.386 [0.386, 0.386] | 0.854 [0.852, 0.856] |

## S6. Whole-archive supplements

Full paired MSE cells, group and region sensitivities, unavailable cases and source calibration support are in the review-response results. Forward task calibration ends before 2010. Normal-spacing tail extrapolation is compared to constant tails under the same observed-window constraint; its outer distribution is assumed, not identified by the seven stored quantiles. Auxiliary rank coupling is not a posterior for the annual estimator.

## S7. Matched numerical release within each archive

After the original release and trend results were inspected, we fixed a further descriptive comparison on the same 3,320 archive settings. In each archive the three-output rule's number of numerical releases was matched exactly by choosing the lowest available raw classifier scores; equal scores were broken by year, and the existing detected-quality exclusion was applied to both rules. The choice uses the target archive's score distribution and its original release count, but no reference outcomes. It is therefore transductive and not a prospective release threshold. The same observed years and D2 estimates are used, while the selected years can differ. Nonselected years retain a visible lower bound if allowed by the same quality flag, and otherwise remain unknown. An unassessed quality diagnostic does not count as a detected flag in this analysis, as in the original pipeline.

Table S15 gives annual errors and working-likelihood trend RMSE; lower-bound counts are not an assertion that the same years or bound values were retained. The number of numerical outputs is matched within each archive, but equal counts do not establish equal total information or comparable prospective risks. Repeated masks are not independent climates. A connected-group bootstrap of paired squared-error differences for forward

**Table S9.** The same 35 observation-only complete-year witnesses under each screening rule. The accepted-record comparison uses observed windows, not unknown complete H30 values. The 49-day and July–August rows are contextual proxies, not published H30 rules.

| Rule | Witnesses | Admitted | Excluded | Above accepted |
|---|---|---|---|---|
| Complete | 35 | 0 | 35 | 35 |
| Annual 95 | 35 | 31 | 4 | 3 |
| Station warm 90 | 35 | 32 | 3 | 2 |
| Fixed warm 90 | 35 | 32 | 3 | 2 |
| July–August | 35 | 30 | 5 | 3 |
| Gap $\leq$49 d | 35 | 34 | 1 | 1 |

**Table S10.** Seven-station stress comparisons on common available station/repetition rows. Heat-associated masks and numbers of missing years differ between arms; unavailable cases remain in the eligible denominator. Coverage and width are supplied in the result table.

| Scenario | Eligible | Available | Base RMSE | Stress RMSE |
|---|---|---|---|---|
| Donors / forward | 30 | 30 | 0.079 | 0.116 |
| Donors / offline | 35 | 35 | 0.037 | 0.047 |
| Heat / forward | 30 | 30 | 0.079 | 0.088 |
| Heat / offline | 35 | 30 | 0.035 | 0.058 |

**Table S11.** Release followed by finite-record trend inference for controlled gaps. The values-and-bounds interval is a Gaussian working-likelihood interval, including regression variability; it is not calibrated missing-data uncertainty.

| Mode | Released data | Eligible | Available | RMSE | Working width |
|---|---|---|---|---|---|
| forward | Values + bounds | 820 | 820 | 0.202 | 3.262 |
| forward | Values | 820 | 820 | 0.210 | – |
| offline | Values + bounds | 840 | 840 | 0.040 | 1.986 |
| offline | Values | 840 | 840 | 0.043 | – |

**Table S12.** Task calibration and independent-draw intervals on exactly the same supported cases. Coverage is a fraction; missing source support is reported separately.

| Setting | Method | $n$ | Original coverage | Task coverage | Width ratio |
|---|---|---|---|---|---|
| calendar / forward | D2 | 40 | 0.925 | 0.975 | 0.947 |
| calendar / forward | Stochastic | 38 | 1.000 | 1.000 | 0.773 |
| calendar / offline | D2 | 95 | 0.937 | 0.958 | 1.167 |
| calendar / offline | Stochastic | 88 | 0.977 | 0.989 | 1.225 |
| crossed / forward | D2 | 39 | 1.000 | 1.000 | 1.112 |
| crossed / forward | Stochastic | 38 | 1.000 | 1.000 | 1.015 |
| crossed / offline | D2 | 301 | 0.977 | 0.980 | 1.236 |
| crossed / offline | Stochastic | 307 | 0.948 | 0.951 | 1.108 |

**Table S13.** Task calibration support. Empirical calibration from small source pools does not establish a finite-sample guarantee.

| Family | Time setting | Estimator | Availability | Minimum source groups |
|---|---|---|---|---|
| calendar | forward | D2 | 0.049 | 3 |
| calendar | forward | stochastic_calibrated | 0.046 | 3 |
| calendar | offline | D2 | 0.113 | 3 |
| calendar | offline | stochastic_calibrated | 0.105 | 3 |
| crossed | forward | D2 | 0.048 | 3 |
| crossed | forward | stochastic_calibrated | 0.046 | 3 |
| crossed | offline | D2 | 0.358 | 3 |
| crossed | offline | stochastic_calibrated | 0.365 | 3 |

**Table S14.** Additional observation-only witnesses from approved USGS daily means. Every candidate remains in the qualification table. Sparse complete-reference support precludes an incidence or transfer inference.

| Station | Year | Complete $n$ | Observed $L$ | Complete max | Excess |
|---|---|---|---|---|---|
| 08377900 | 2016 | 3 | 14.990 | 14.853 | 0.137 |
| 08377900 | 2018 | 3 | 15.537 | 14.853 | 0.683 |
| 08377900 | 2020 | 3 | 15.737 | 14.853 | 0.883 |
| 08405200 | 2013 | 2 | 28.693 | 28.327 | 0.367 |
| 08405200 | 2015 | 2 | 28.703 | 28.327 | 0.377 |
| 08405200 | 2016 | 2 | 28.537 | 28.327 | 0.210 |
| 08405200 | 2019 | 2 | 28.833 | 28.327 | 0.507 |
| 08405200 | 2020 | 2 | 28.970 | 28.327 | 0.643 |
| 07364133 | 2015 | 1 | 29.427 | 28.767 | 0.660 |

controlled gaps gives [-0.0017, 0.0072] (°C/decade)$^2$; this is descriptive and spans zero. The likelihood treats released predictions as exact, censoring may be informative, and its intervals are not calibrated missingness-only uncertainty. No model was refitted for this comparison.

**Table S15.** Per-archive matched numerical output. Annual errors exceed 0.5 °C. Counts and RMSE are original three-output rule / score-only comparator; RMSE uses common converged release-plus-bound working-likelihood fits and units °C per decade.

| Scenario | Numeric | Errors | Bounds | Fits | RMSE |
|---|---|---|---|---|---|
| Calendar, offline | 6,345 | 11 / 9 | 590 / 590 | 840 | 0.0198 / 0.0237 |
| Calendar, forward | 3,435 | 43 / 32 | 371 / 370 | 819 | 0.1240 / 0.1220 |
| Controlled, offline | 5,543 | 242 / 229 | 1378 / 1378 | 840 | 0.0405 / 0.0407 |
| Controlled, forward | 3,174 | 397 / 370 | 603 / 603 | 820 | 0.2024 / 0.1959 |

The audit separately reconstructed the selection masks for all 3,320 archives using Python ordering, recomputed 3,319 available score-comparator numerical-only OLS slopes and checked all eight summary cells. A second implementation refitted all 6,640 rule-by-archive censored likelihoods using normal log-density and survival functions. Convergence status agreed for every fit; 6,638 converged in both implementations, and their largest slope difference was approximately $1.2 \times 10^{-6}$ °C per decade. This numerical check used the same observations, selected cases and working likelihood; it did not independently reconstruct source sensors or models, establish likelihood validity or verify interval standard errors.

## S8. Expanded missingness-mechanism sensitivities

After the earlier results and seven-site pilot had been inspected, we fixed an expansion to the first five masks for all 41 forward-eligible original archive stations in 17 connected groups. Only controlled gaps and one seed-zero source-model assignment are studied. For the donor sensitivity, the target masks are unchanged and those same dates are hidden in every other panel column before local fitting. All 205 settings have at least one finite donor-panel observation removed; the total of 427,375 finite donor-panel cells counts repeated masks and may include columns not ultimately selected by the local estimator. It is not a count of independent outages or necessarily of observations the estimator would have used.

For the heat-associated sensitivity, the original 35% Bernoulli draw determines the number of masked years in each station–repetition archive. An independent seeded exponential-race sample chooses exactly that number of eligible years, with weight five for years whose annual air H30 is at least the first-seven-year 80th percentile, and weight one otherwise. Air information determines a synthetic test mask; it is not a hidden water-temperature outcome. The full cohort has 943 masked years in either design, of which 511 and 700 are classified as air-hot. However, the selected years have different existing within-year gap shapes and positions, so the total hidden target days change from 55,950 to 57,540. These paired scenarios cannot isolate the causal effect of heat association alone or represent a measured outage mechanism.

Table S16 compares fixed-prediction cohorts for both estimators, with availability reported on the common subset. Nominal trend interval coverage, widths and scores use independent annual ranks and the original normal-spacing tails; the intervals are not

validated climate-transfer guarantees. Connected-group resampling of paired D2 squared-error differences gives [0.0030, 0.0434] for synchronized donors and [-0.0040, 0.0091] for heat-associated masking, in $(°C/decade)^2$. These describe a selected archive and depend on group exchangeability. Removing Bavaria without refitting changes baseline and donor RMSE to 0.0829 and 0.1287, and baseline and heat RMSE to 0.0829 and 0.0816. Other fixed-prediction leave-region-out summaries are supplied in the aggregate files. Repeated masks are not independent climates; original forward source roles are not uniformly strict named-region holdouts.

**Table S16.** Expanded paired controlled-gap stresses, original / stress. RMSE and interval widths are °C per decade. Coverage is for nominal 95% trend intervals; scores are also in °C per decade. Common fitted cohorts differ between estimators and scenarios.

| Scenario | Estimator | Cases | RMSE | Coverage | Width | Score |
|---|---|---|---|---|---|---|
| Donor gaps | D2 | 205 | 0.1555 / 0.2024 | 92.2 / 86.8 | 0.465 / 0.479 | 0.756 / 1.039 |
| Donor gaps | Stochastic | 190 | 0.1386 / 0.1877 | 87.4 / 76.3 | 0.404 / 0.418 | 0.808 / 1.257 |
| Heat, fixed years | D2 | 205 | 0.1555 / 0.1655 | 92.2 / 88.8 | 0.465 / 0.456 | 0.756 / 0.806 |
| Heat, fixed years | Stochastic | 187 | 0.1370 / 0.1552 | 87.7 / 87.2 | 0.403 / 0.383 | 0.776 / 0.884 |

An independent audit reconstructs all 205 baseline masks, 205 synchronized masks and 205 heat-associated masks, and recomputes eight paired RMSE and mean-squared-error cells. It uses the same source observations, random seeds and saved fitted predictions; it does not independently refit the models, check bootstrap intervals or establish sensor quality.

## S9. Heat-associated years with matched gap templates

After examining earlier results and the available gap templates, we repeated the forward controlled-gap sensitivity for 41 stations and five archive masks per station. The selected heat-associated years are the same as in the prior heat sensitivity. For years masked in both the baseline and heat design, the original gap is retained; remaining baseline offset–length templates are transferred to the newly selected heat years. Every archive retains its original masked-year count, multiset of templates and actual number of hidden days. Total hidden days are 55,950 in either design, and air-hot masked years increase from 511 to 700; repeated masks are not independent observations. Offsets are relative to the year, and the relationship of a transferred gap to a particular seasonal peak can change.

The matched-gap plan incorrectly described the first-seven-year air baseline as entirely pre-2010. In 4 of the 41 stations, one of those seven reference-year labels is 2010 or later. The saved implementation uses the first seven reference years; no predictions were changed by this clarification. Air information constructs a synthetic retrospective mask and is not a hidden water-temperature outcome, but the mechanism is not a strictly prior-vintage or prospective outage generator.

After viewing the pairwise results, we additionally restricted all three designs and both methods to the same 183 available station–replicate settings from 41 stations and 17 connected groups (Table S17). Group-resampled differences in paired MSE for matched minus baseline are [-0.0096, 0.0039] for D2 and [-0.0074, 0.0081] for the stochastic comparator, in $(°C/\text{decade})^2$. The shared cohort is post-hoc, and these differences change in group-equal weighting. The fixed heat-year count and gap-template distribution do not isolate year,

position and peak alignment, establish a natural-outage mechanism, or prove climate transfer.

**Table S17.** Three-way common cohort, baseline / previous heat / matched heat, with independent annual ranks. RMSE, widths and interval scores are °C per decade; coverage is for nominal 95% trend intervals.

| Method | RMSE | Coverage (%) | Width | Interval score |
|---|---|---|---|---|
| D2 | 0.153 / 0.162 / 0.142 | 92.3 / 90.2 / 91.3 | 0.464 / 0.445 / 0.445 | 0.743 / 0.770 / 0.696 |
| Stochastic | 0.138 / 0.155 / 0.136 | 88.0 / 87.4 / 85.8 | 0.408 / 0.385 / 0.391 | 0.789 / 0.878 / 0.865 |

Separate checks reconstruct all 205 masks from the same underlying panels and seeded selection and independently recompute 32 paired summary rows, 96 fixed-prediction leave-region-out rows, 24 three-way summary rows and 16 three-way contrasts, including seeded group resampling. The underlying observations, trained source models and local daily refits are reused; these checks do not constitute independent physical or climate validation.

## S10. Source-model sensitivity of matched heat-gap archives

After observing the earlier source-role and matched-gap results, we held each target mask and daily feature record fixed and re-applied three source-model choices. The original version uses the original source model in every region. The strict-original version replaces only the affected HUC17 model with its earlier refit excluding all HUC17 source groups; elsewhere it is identical to the original. The candidate-provider version uses the earlier target-region-excluded models, which also change the provider set, sample size and role allocation. In all, 205 candidate and 15 strict-original matched-gap settings were

re-scored. The source models and daily feature fits were not independently retrained for this comparison.

Table S18 uses the same 184 archives at 41 stations and 17 connected groups, where all six source-version by baseline/stress results are available for both methods. The nominal 95% trend intervals use independent annual ranks. Group resamples concern repeated masks and depend on the sampled groups; all displayed paired-MSE intervals cross zero. Changing source versions also changes group-equal estimates, so the apparent average direction is not a general effect of heat-associated outage. The four stations whose seven-year air reference includes an evaluation-period year, template transfer and retrospective scenario selection remain limitations of the same stress design.

**Table S18.** Matched-gap source sensitivity, baseline / stress on the same two-method, six-way common cohort. RMSE is °C per decade; group-equal paired MSE difference and group-resampled intervals are $(°C/\text{decade})^2$.

| Method | Source | RMSE | Coverage (%) | Group-equal MSE | Paired MSE interval |
|---|---|---|---|---|---|
| D2 | Original | 0.152 / 0.142 | 92.4 / 91.3 | 0.0013 | [-0.0092, 0.0039] |
| D2 | Strict original | 0.153 / 0.142 | 92.4 / 90.8 | 0.0008 | [-0.0092, 0.0036] |
| D2 | Candidate providers | 0.153 / 0.134 | 88.0 / 90.8 | -0.0013 | [-0.0130, 0.0026] |
| Stochastic | Original | 0.138 / 0.136 | 88.0 / 85.9 | 0.0057 | [-0.0071, 0.0081] |
| Stochastic | Strict original | 0.138 / 0.136 | 88.0 / 85.9 | 0.0057 | [-0.0070, 0.0087] |
| Stochastic | Candidate providers | 0.145 / 0.140 | 87.0 / 85.9 | 0.0042 | [-0.0084, 0.0067] |

After inspecting the source-specific results, we also computed a directly paired difference between the stress-minus-baseline MSE changes under the candidate and original sources. For D2 with independent ranks, this difference is -0.0024 with group-resampled interval

[-0.0052, -0.0001] (°C/decade)$^2$. The interval does not cross zero but its upper limit is close to zero, whereas each within-source interval in Table S18 crosses zero; paired contrasts account for covariance and are not equivalent to comparing two separate intervals. The preliminary calculation used separate 4,000-draw streams across rank representations that shared the same point predictions, and its limits differed near zero. After inspecting those results, we retained them and used 40,000 shared group resamples per method and source comparison. In a further post-hoc fixed-prediction check, deleting one connected group and recomputing the interval makes it cross or touch zero for 6 of 17 possible deletions. This is one of 16 reported post-hoc cells, with no multiplicity adjustment; rank representations share point estimates and are not independent tests. Candidate source membership, role assignments and sample size change together. It is evidence of sensitivity in these selected archives, not a causal provider or climate-transfer effect.

Independent checks compare every non-source-model column in 220 replay files (1,003 annual rows), confirm that strict and original predictions coincide outside HUC17, and reconstruct 48 summary cells, 144 fixed-prediction leave-region-out cells, all 16 paired source interactions and 68 leave-one-group resamples from saved predictions. These are numerical checks on shared observations and fitted models, not an independent climate, sensor or legal-rights evaluation. The first attempt to distribute tasks to eight processes failed to deserialize a task; the same plan completed serially with receipt checks.

## S11. Source-model sensitivity of expanded stresses

Following inspection of the original source-role audit and expanded stress results, we replayed the stressed daily features and masks through two already fitted alternative source-model sets. The strict-original version replaces only the forward HUC17 model,

the sole affected target region in this cohort, with the model refitted after excluding every HUC17 connected group from its source roles; other source assignments remain as before. This removes source cases and changes source sample size. The candidate version replaces the models for all six forward target regions with the separately refitted candidate-provider models, which exclude all target-region groups but also change source providers, source sample size and role assignments. Source-model outputs and calibration offsets are recalculated; the previously fitted local daily features and target masks remain fixed. Neither comparison is prospective or a clean causal experiment for a single change.

Table S19 uses, separately for each estimator and stress, only archives available under both baseline and stress in all three source versions. The independent annual-rank intervals use the same six-way common interval cohort, with the same normal-spacing tail assumptions. D2 has 205 common archives; the stochastic method has 190 for donor gaps and 187 for the fixed-count heat stress. Repeated masks are not independent climates. Descriptive connected-group bootstrap and fixed-prediction region deletion results accompany the aggregate files. In the heat scenario the number of masked years is fixed within archive, but the number and placement of hidden days can change; in the donor scenario target masks are identical. These simulations do not verify a natural outage mechanism or transfer to new climates.

The source-role audit checked the positive-weight roles of the original and alternative models. All 440 alternative stress reanalyses preserved the target masks, hidden-value checks, plug-in values and observed lower bounds. The audit checked all 205 candidate baseline receipts against their source models, calibration, original features and saved outputs, and replayed the 15 affected strict-original baselines, including 60 annual predictions

**Table S19.** Source-model sensitivity on six-way common archives, baseline / stress. RMSE, width and score use °C per decade; coverage is the nominal 95% trend interval percentage. Strict changes only the affected HUC17 source model; candidate changes source providers and assignments across all regions.

| Scenario | Estimator | Source | Cases | RMSE | Coverage | Width | Score |
|---|---|---|---|---|---|---|---|
| Donor gaps | D2 | Original | 205 | 0.1555 / 0.2024 | 92.2 / 86.8 | 0.465 / 0.479 | 0.756 / 1.039 |
| Donor gaps | D2 | Strict | 205 | 0.1561 / 0.2027 | 92.2 / 86.8 | 0.467 / 0.475 | 0.758 / 1.050 |
| Donor gaps | D2 | Candidate | 205 | 0.1514 / 0.2068 | 88.8 / 82.4 | 0.425 / 0.502 | 0.738 / 1.117 |
| Donor gaps | Stochastic | Original | 190 | 0.1386 / 0.1877 | 87.4 / 76.3 | 0.404 / 0.418 | 0.808 / 1.257 |
| Donor gaps | Stochastic | Strict | 190 | 0.1386 / 0.1877 | 87.4 / 76.3 | 0.404 / 0.418 | 0.808 / 1.257 |
| Donor gaps | Stochastic | Candidate | 190 | 0.1455 / 0.1969 | 86.8 / 76.3 | 0.392 / 0.461 | 0.799 / 1.364 |
| Heat, fixed years | D2 | Original | 205 | 0.1555 / 0.1655 | 92.2 / 88.8 | 0.465 / 0.456 | 0.756 / 0.806 |
| Heat, fixed years | D2 | Strict | 205 | 0.1561 / 0.1661 | 92.2 / 88.3 | 0.467 / 0.457 | 0.758 / 0.812 |
| Heat, fixed years | D2 | Candidate | 205 | 0.1514 / 0.1642 | 88.8 / 86.8 | 0.425 / 0.410 | 0.738 / 0.877 |
| Heat, fixed years | Stochastic | Original | 187 | 0.1370 / 0.1552 | 87.7 / 87.2 | 0.403 / 0.383 | 0.776 / 0.884 |
| Heat, fixed years | Stochastic | Strict | 187 | 0.1370 / 0.1552 | 87.7 / 87.2 | 0.403 / 0.383 | 0.776 / 0.884 |
| Heat, fixed years | Stochastic | Candidate | 187 | 0.1443 / 0.1677 | 87.7 / 82.9 | 0.390 / 0.373 | 0.764 / 0.947 |

and 180 trend rows. A second implementation independently recomputed the six-way common cohorts, 24 aggregate cells, 144 region-deletion cells and 4,000 group bootstrap draws per cell from 9,840 saved case rows. These checks did not independently refit source models, check sensor accuracy or validate climate transfer.

## S12. Exploratory projected-point failure score

After inspecting prior projection and risk results, we fitted six region-specific binary classifiers to identify absolute D2 point error above 0.5 °C after the observed-bound

projection. For each target region, the source estimator, interval-calibration, risk-learning and rule-calibration connected-group roles are disjoint, all groups appearing in the target region are excluded, and forward source cases precede 2010. Training uses the assigned risk-learning source groups, weighted equally by connected group. The new classifier uses the existing decision-time feature matrix, the projected 90% interval width and the projected point's distance above the visible bound; neither the old risk score nor the held reference is a predictor. Each model uses 120 low-leaf-count boosted trees with the fixed settings documented in the analysis plan. Six to eight source training groups contribute per model.

For each region and mask family, separate source rule-calibration groups select an old-score and a new-score cutoff to approximate 80% group-equal-weight numerical retention on valid source cases, without using failure labels to choose the cutoff. Both cutoffs are fixed before target scoring. Target numerical retention can differ. A second, transductive comparison ranks the new target scores and matches the old-score numerical count separately within each target region and family; ties are broken deterministically without target outcomes. It is not a prospective cutoff, and equal counts need not retain the same years or bound values.

The candidate annual caches contained no reference H30 values. Therefore source models, source cutoffs, target scores and selections were saved first; the target reference was then joined by station, masked year, family and replicate from the separately saved complete-archive record. A second comparison against the original source station–year–family H30 verified all 7,583 reference values, and the saved selections did not change. These target

archives and source data had nevertheless been inspected in earlier analyses; this is not an unopened reference experiment.

Table S20 gives pooled forward results. Source thresholds in the controlled-gap family give old/new numerical retention 89.6% / 85.3% and selected failure 10.72% / 9.50%. At the matched 3,385 outputs, the original and projected-risk scores have 363 and 383 failures, respectively; the connected-group bootstrap interval for the difference in selected failure rates (new minus old) is [-0.73, 1.86] percentage points. Group resampling is descriptive and repeated masks are not independent climates. The old classifier targeted an unprojected point; the new classifier targets the projected point, but a better-aligned training label does not itself imply better deployment decisions. The source masks degrade one year at a time, while target archives are jointly degraded across years. These results do not establish calibrated probabilities, group-level risk control, natural-outage performance or independent-climate transfer.

**Table S20.** Forward annual failure comparison, old raw score / projected-point score. An error exceeds 0.5 °C. Source-threshold counts can differ; the target count match is transductive and uses no target outcomes.

| Mask | Selection | Cases | Numerical | Failures | Failure (%) |
|---|---|---|---|---|---|
| Calendar | Source threshold | 3,806 | 3,431 / 3,112 | 6 / 6 | 0.17 / 0.19 |
| Calendar | Target count match | 3,806 | 3,431 / 3,431 | 6 / 6 | 0.17 / 0.17 |
| Controlled | Source threshold | 3,777 | 3,385 / 3,222 | 363 / 306 | 10.72 / 9.50 |
| Controlled | Target count match | 3,777 | 3,385 / 3,385 | 363 / 383 | 10.72 / 11.31 |

The independent audit refits the six classifiers using the same source records and feature implementation, rechecks the disjoint roles, reproduces the twelve family-specific source threshold pairs, reconstructs all target scores and label-free selections, and recomputes group bootstrap and region-deletion summaries. It also checks reference attachment against both saved reference sources. It does not independently reconstruct physical observations, establish source permissions or verify the classifier's prospective reliability.

## S13. Seasonal-window and screening-rule diagnostics

The previously selected source sample contains 96 eligible long-record stations; this calculation uses the 95-station project-processed subset with a documented candidate redistribution path. None of the 35 recorded witnesses is at the excluded LUBW station, although it remains part of the original eligible-station denominator. The subset comprises 2,122 complete station-years at 95 stations and 33 connected groups. For each complete year, we compare the full-year reported H30 with the largest 30-day mean whose every day lies from 1 July through 31 August. Of those years, 779 exceed the summer-window value by more than the numerical tolerance $5 \times 10^{-6}$ °C, and 304 exceed it by more than 0.5 °C. The mean difference is 0.255 °C. These are complete recorded years, not simulated reconstruction errors; neither the tolerance nor the 0.5 °C threshold is an estimate of sensor error.

The comparison was specified after the original witnesses and literature had been examined. It is not a reproduction of Beaufort et al.'s multiyear thermal peak or their summer air-temperature climate correction; that study also excluded sites with strong human influence. The present sample contains regulated and incompletely characterized

sites, covers different locations and years, and the fully-contained July–August window differs from a peak-center criterion. Consequently our fraction must not be compared directly with their reported fraction. The incomplete-year comparison is also conditional on the 35 already selected record witnesses: all have a fully observed July–August 30-day window, and 9 have a larger fully observed 30-day window elsewhere in the year. This does not establish the full H30 of an incomplete year or a population prevalence.

Table S21 counts previously eligible station-years rejected by four explicit study rules, including how many still supply at least one fully observed 30-day window. A lower bound is information about the reported record, not a complete annual estimate or a guarantee of better trends. The 90% station-specific warm-season rule uses a leave-target-year-out climatology; the complete July–August row is a contextual comparator rather than the full published climate-correction procedure. The records are not an independent validation sample and no sensor error has been estimated.

**Table S21.** Study screening rules on 2,656 eligible station-years in the 95-station candidate subset. Rejected-with-bound means the rejected incomplete year contains at least one complete observed 30-day window.

| Rule | Accepted | Rejected | Unknown | Rejected with bound | Stations |
|---|---|---|---|---|---|
| Complete year | 2,122 | 534 | 0 | 518 | 70 |
| Annual 95% | 2,550 | 106 | 0 | 90 | 48 |
| Station warm season 90% | 2,590 | 66 | 0 | 50 | 30 |
| Complete July–August | 2,472 | 184 | 0 | 168 | 58 |

After inspecting the largest annual gaps, we separately summarized existing project regulation labels and removed the two sites with the largest individual gaps (Table S22). The 2010 USGS analysis identified Blue River and Fall Creek as dam-affected monitoring sites in 2001–2002; it does not establish the cause or measurement quality of every year examined here. The existing *unmatched gages* label does not mean unregulated, and labels and sites were not randomized. Removing these two post-hoc sites leaves 749 of 2,086 complete station-years above the numerical tolerance, with mean difference 0.197 °C. These summaries do not identify causal dam effects or transfer to streams absent from the sample.

**Table S22.** Post-hoc seasonal-window influence by existing project label and after removing two selected sites. "Above" is greater than $5 \times 10^{-6}$ °C; the second count exceeds 0.5 °C. Labels are observational metadata, not independently verified treatments.

| Cohort | Stations | Years | Above | $> 0.5$ | Mean (°C) |
|---|---|---|---|---|---|
| All stations | 95 | 2,122 | 779 | 304 | 0.255 |
| Exclude two selected sites | 93 | 2,086 | 749 | 279 | 0.197 |
| Label: regulated | 23 | 397 | 173 | 103 | 0.725 |
| Label: unmatched gages | 61 | 1,542 | 569 | 196 | 0.158 |
| Label: unknown | 11 | 183 | 37 | 5 | 0.049 |

The comparison and screening totals were independently reconstructed using pandas rolling windows on the same snapshot. The post-hoc influence totals and 36 selected-site

peak-window dates were independently checked using NumPy convolution; the stored regulation labels were checked against the project's source metadata. Neither audit verifies the original provider's measurements or whether each project regulation label is accurate. The supplied 95-station snapshot permits the window and screening arithmetic to be rerun, subject to its attribution and reuse conditions.

## S14. Literature-inspired interannual thermal-peak comparison

After inspecting earlier results and data availability, we selected all 22 of the previously eligible 95 stations with complete reported water years from 2009 through 2017. They belong to 10 connected groups; project labels mark one as regulated and leave the others unmatched to the regulation inventory, not necessarily natural. Following the structure described by Beaufort et al. (`https://doi.org/10.5194/hess-26-3477-2022`), we examine a nine-year mean of annual 30-day peaks and fill missing summer daily water with station-specific regressions on right-aligned 2–10-day air-temperature means. In each synthetic experiment, we hide 1–8 whole years in every possible combination, select the lag with the highest in-sample $R^2$ using retained summers only, and fit the regression without withheld-year water. We calculate the 30-day maximum wholly contained in July–August for observed and predicted years. All nine years hidden is unavailable because no local fitting data remain. The 11,220 station–mask cases reuse only 22 stations and are not independent climate replicates.

The primary reference is the mean of the nine observed July–August 30-day maxima. A separately reported reference is the mean of the nine full-calendar-year H30 values; this changes the target, so any summer-only estimator may omit a peak outside July–August. The unfilled comparison averages only the retained years; the filled comparison averages

retained observed peaks and reconstructed missing-year summer peaks. Both methods are evaluated on the same cases. These are point estimates of interannual averages, not annual H30 recovery, trend, interval calibration or release-rule validation. The observed-only mean bias across the exhaustive symmetric masks is zero for the summer reference by construction; this does not establish unbiasedness under real missingness.

Table S23 reports station-equal MAE and 4,000 connected-group bootstrap intervals for the filled-minus-observed absolute-error difference, with one row per station in the group resample. The comparison is post-hoc and uses ten project connected groups; the resampling interval is descriptive and depends on those groups. Other missing-year counts and leave-one-provider-out results are in the aggregate output. Removing a provider does not refit models or establish independent climates.

This is an adaptation, not a reproduction of the French study: it uses a different station population, processed water records, NASA POWER-based daily air instead of SAFRAN, explicit train-only fitting, all combinations of hidden years, and a July–August-contained peak definition. The published article does not establish all of our implementation choices. NASA POWER's gridded local-solar air days may not align with provider water days; the original API version and raw response for these 22 cached stations were not retained. After viewing the proxy results, we requested current NASA POWER responses for 15 representatives of distinct stored air series. All 9,585 comparable summer and lag-input days matched within $10^{-5}$ °C. This does not establish historical version, unrequested stations, air or water accuracy, clock alignment or independent-climate generality. A separate implementation recomputed 11,220 masks and all 16 summary cells from the same processed inputs; it did not reproduce the original authors' datasets or code.

**Table S23.** Adapted summer-regression proxy against two distinct nine-year targets. MAE and filled-minus-observed MAE differences are °C; cases are repeated year masks.

| Reference | Hidden years | Cases | Observed MAE | Filled MAE | Difference | Group interval |
|---|---|---|---|---|---|---|
| Summer | 1 | 198 | 0.100 | 0.069 | -0.031 | [-0.051, -0.020] |
| Summer | 2 | 792 | 0.156 | 0.112 | -0.043 | [-0.070, -0.028] |
| Summer | 4 | 2772 | 0.266 | 0.189 | -0.078 | [-0.114, -0.057] |
| Summer | 6 | 1848 | 0.418 | 0.285 | -0.133 | [-0.184, -0.103] |
| Summer | 8 | 198 | 0.803 | 0.530 | -0.273 | [-0.416, -0.192] |
| Full-year | 1 | 198 | 0.190 | 0.192 | 0.002 | [-0.016, 0.012] |
| Full-year | 2 | 792 | 0.222 | 0.223 | 0.001 | [-0.030, 0.018] |
| Full-year | 4 | 2772 | 0.305 | 0.288 | -0.017 | [-0.068, 0.012] |
| Full-year | 6 | 1848 | 0.437 | 0.370 | -0.067 | [-0.143, -0.023] |
| Full-year | 8 | 198 | 0.805 | 0.581 | -0.224 | [-0.390, -0.127] |

## S15. Direct archive comparison with a literature-inspired summer air proxy

After viewing the original archive and interannual-proxy results, we evaluated a different, deterministic adaptation on the same 3,320 jointly masked archives. For each target year, all target-archive gaps are hidden before fitting, the entire target year is excluded from the local summer water–air fit, and forward fits exclude water from 2010 onward. We require at least three complete eligible training years, 30 usable summer days from at least three years, and common finite predictors for all 2–10-day right-aligned air averages; the lag maximizing in-sample $R^2$ is selected. We fill missing July–August water days and take the maximum of the reconstructed summer 30-day peak and the fully observed full-year

30-day bound. When summer contains no gaps, the result is the observed full-year bound alone. Missing air and unsupported fits remain unavailable. This differs in data, target and implementation from Beaufort et al. (https://doi.org/10.5194/hess-26-3477-2022); it is not their code or a test of their original multiyear thermal-peak estimator.

There are 21,446 masked station-year records in the repeated archive settings; 21,278 proxy outputs are available. Of all records, 11,334 have no summer gaps and return only the observed bound; 14,784 available outputs equal that bound, including cases where a regression was fitted. A bound can remain below the true annual maximum when the missing peak is outside July–August. These are repeated synthetic masks, not independent site-years or climates. The source roles, pre-existing fixed models, air-day alignment and restricted-data access limitations remain. The proxy supplies no predictive interval. Annual retrospective strata and pairwise availability are reported in the aggregate output; the bound-equals-reference stratum uses the hidden reference and cannot be selected at deployment.

Table S24 uses cases where all four point methods are finite. Trend rows exclude archives with no masked year and use the same finite-record OLS target. Differences are descriptive after selecting this analysis; resampling connected groups and fixed-prediction leave-one-region-out summaries do not establish out-of-region prediction or causal superiority. Independent least-squares and rolling-window code checked 52 selected archives in seven regions, 310 masked years and target-year fit perturbations, plus the reported aggregate cells. It used the same processed sources and fixed baseline predictions.

**Table S24.** Common-cohort annual and nontrivial-archive point comparisons. Annual MAE is °C; trend RMSE is °C per decade. Counts are repeated mask records or archives.

| Setting | Annual $n$ | D2 MAE | Proxy MAE | Archive $n$ | D2 RMSE | Proxy RMSE |
|---|---|---|---|---|---|---|
| Calendar, offline | 6,891 | 0.018 | 0.037 | 822 | 0.009 | 0.018 |
| Calendar, forward | 3,676 | 0.029 | 0.047 | 699 | 0.062 | 0.083 |
| Long gap, offline | 6,786 | 0.165 | 0.294 | 818 | 0.028 | 0.068 |
| Long gap, forward | 3,683 | 0.225 | 0.361 | 736 | 0.164 | 0.237 |

## S16. Past-only warm-season sensitivity

The original station-specific warm season leaves out the evaluated year but can use observations from later years. After inspecting that comparison and the 35 selected witnesses, we recomputed the same 15-day-smoothed day-of-year climatology and five-month selection with only daily values from years strictly before each evaluated year. The source is the current processed archive, not a historically timestamped data vintage. The 95 stations and 35 witnesses were themselves selected using records available after some evaluated years, so this is a retrospective information-separation sensitivity, not a prospective site recruitment or field performance test.

Two past-only variants are reported. One uses any earlier finite water observation, which can amount to only 1 day for an incomplete year; climatological interpolation with so little data is not evidence of adequate support. The other requires at least four fully observed prior calendar years before classifying an incomplete year. Complete years are accepted even if the climatology is unavailable, because every possible warm season is fully observed. The four-year threshold is an exploratory study choice, not a published

standard. Unsupported incomplete years remain unknown rather than being counted as rejections. On 2,656 eligible station-years, the past-only variant without a minimum changes 1 decision relative to the original leave-year-out rule; the four-year version leaves 127 years unknown, 122 of which still have an observed 30-day lower bound.

Table S25 also compares the reported observed-window maximum in each rejected year to the greatest reported window from earlier years accepted by that same rule. A rejected year contributes only if both its observed bound and an earlier accepted reported window exist. Earlier accepted incomplete years can have an unobserved higher H30, and these are not comparisons with physical sensor truth. The leave-year-out rule's earlier classifications may use years later than their evaluation year; only the past-only rules enforce the stated temporal information boundary. Among the 35 retrospectively selected witnesses, the four-prior-complete-years variant accepts 27, rejects 2 and leaves 6 unknown; 2 rejected witnesses exceed the earlier accepted reported-window maximum. These counts are not an incidence estimate.

**Table S25.** Retrospective screening in the 95-station candidate subset. "Above prior" counts rejected years whose observed 30-day window exceeds all earlier accepted reported windows. Witness column gives accepted / rejected / unknown among 35 previously selected incomplete years.

| Rule | Accepted | Rejected | Unknown | Rejected with bound | Above prior | Witnesses |
|---|---|---|---|---|---|---|
| Complete year | 2,122 | 534 | 0 | 518 | 86 | 0 / 35 / 0 |
| Annual 95% | 2,550 | 106 | 0 | 90 | 11 | 31 / 4 / 0 |
| Warm 90%, leave year out | 2,590 | 66 | 0 | 50 | 9 | 32 / 3 / 0 |
| Warm 90%, past data | 2,589 | 67 | 0 | 51 | 9 | 32 / 3 / 0 |
| Warm 90%, past + 4 complete | 2,487 | 42 | 127 | 31 | 3 | 27 / 2 / 6 |

An independent implementation rebuilt all 2,656 earlier-only climatologies, 13,280 rule states and prior-record comparisons on the same processed daily records. In 35 witness-year probes, shifting the target and all later observations by 5 °C did not change the earlier-only warm season. These checks do not verify historical data vintages, station selection, provider quality control or physical measurement accuracy.

## S17. Conservative source-group maximum sensitivity

After earlier annual, trend and classifier results had been inspected, we combined the candidate source model's assigned risk-learning and rule-calibration connected groups for each target region. These groups are disjoint from the numeric quantile estimator and its interval-offset groups and exclude groups in the held target region. The risk-learning groups trained a separate failure classifier, but no classifier score or selection is used in this interval analysis. Replacing the stored classifier with a constant score in 300 source-case probes left both methods' seven annual quantiles unchanged. Source cases are before 2010 and use single-year masks. The source and target observations were already opened; this is a post-hoc sensitivity, not a new prospective experiment.

For each valid source group, we calculate the maximum of the nonnegative distances by which a reference falls outside its projected 90% interval. With $G$ groups, the expansion is the $\lceil 0.9(G+1) \rceil$-th ordered group maximum if that rank exists and is positive; otherwise expansion is unsupported. The six forward regions have 12–16 contributing groups per method and family, so the chosen rank is the maximum among the available group maxima. This finite order statistic does not establish a 90% group-coverage guarantee: exchangeability is not demonstrated, source cases use individual-year masks, target archives have joint multiyear gaps, and groups have different sizes and climates. Both methods'

lower endpoints are constrained to the visible observed window. Table S26 compares the same target cases; the hidden-peak subset uses the complete reference and cannot be selected at decision time. All evaluated cases in each of the 17 target groups happen to be covered after expansion in the controlled-gap cohort, but those groups are not independent-climate replications.

**Table S26.** Nominal 90% annual intervals in common forward controlled-gap cases, baseline / source-group-maximum expansion. Width and interval score are °C; lower score is better.

| Method | Stratum | Cases | Coverage (%) | Width | Score |
|---|---|---|---|---|---|
| D2 | All | 3,683 | 89.2 / 100.0 | 0.734 / 4.690 | 1.442 / 4.690 |
| D2 | Hidden peak | 2,146 | 81.9 / 100.0 | 1.161 / 5.429 | 2.349 / 5.429 |
| Stochastic | All | 3,683 | 87.3 / 100.0 | 0.650 / 5.793 | 1.476 / 5.793 |
| Stochastic | Hidden peak | 2,146 | 78.2 / 100.0 | 1.078 / 6.525 | 2.496 / 6.525 |

We then propagated the same fixed source radii through the original masked archives and three rank constructions. Of 820 settings, 82 lack at least one annual distribution and 2 have no masked year; 736 nontrivial settings in 17 connected groups support both methods and all rank constructions. Two hypothetical quantile representations share the same expanded 5th and 95th percentiles: one changes only endpoints, and the other also tapers changes through interior quantiles. They preserve the same point median and use identical ranks, but their interpolated interior and normal-spacing tails differ. Table S27 shows empirical nominal 95% trend coverage, width and interval score. On the 685 archives

with the earlier narrow source correction and all ranks, the independent-rank tapered D2 interval width is 0.403 with that earlier correction and 2.212 °C per decade with group maxima; the corresponding stochastic widths are 0.376 and 2.747. Matching the annual endpoints does not identify the trend distribution, and observed coverage on these selected masks does not certify coverage for future archives.

**Table S27.** Nominal 95% forward trend intervals on 736 common nontrivial archives. Each entry is baseline / endpoint-only / tapered; widths and interval scores are °C per decade.

| Method | Ranks | Coverage (%) | Width | Score |
|---|---|---|---|---|
| D2 | Independent | 88.6 / 100.0 / 100.0 | 0.399 / 3.102 / 2.222 | 0.735 / 3.102 / 2.222 |
| D2 | Shared parameter | 88.7 / 100.0 / 100.0 | 0.395 / 3.035 / 2.178 | 0.732 / 3.035 / 2.178 |
| D2 | Residual blocks | 88.9 / 100.0 / 100.0 | 0.400 / 3.022 / 2.229 | 0.714 / 3.022 / 2.229 |
| Stochastic | Independent | 84.5 / 100.0 / 100.0 | 0.366 / 3.968 / 2.751 | 0.896 / 3.968 / 2.751 |
| Stochastic | Shared parameter | 83.4 / 100.0 / 100.0 | 0.362 / 3.894 / 2.700 | 0.902 / 3.894 / 2.700 |
| Stochastic | Residual blocks | 83.8 / 100.0 / 100.0 | 0.369 / 3.881 / 2.781 | 0.882 / 3.881 / 2.781 |

The independent annual audit reconstructed 88,053 source case–method predictions, 352 source group maxima, 14,714 target case–method scores and the annual summary cells. An isolation check identified a previously implicit pre-2010 bridge-repair table needed for the stochastic source cases. The original source plan did not list this file: applying the saved table reproduces all source group maxima, while omitting it changes source-case availability and three of the twelve stochastic region–family radii; D2 radii are unchanged. The audit now requires the table explicitly and fails if it is absent or changed. A separate calculation reconstructs 13,248 trend treatment rows from saved quantiles and rank arrays

and checks all 685 overlapping earlier rank snapshots. These numerical audits use the same source and target observations and do not validate exchangeability, physical sensors, provider rights or an independent climate.

Further post-hoc replay identified bridge-repair dependencies in the twenty historical source-seed exports and in the candidate-provider archive calibration. Reapplying the twenty saved repairs reproduced the stored D2 and stochastic quantiles on all 2,914,470 seed–case records. Without those repairs, 1,081,252 records lost complete stochastic quantiles and 1,018,696 records with complete quantiles in both replays had at least one changed quantile. These counts include repeated masks and are not independent station-years. In the archive calibration, applying the two saved repairs reproduced all 26 mode–region–method offset sets; omitting repairs changed offsets in all 13 stochastic sets and none of the 13 state sets. The original candidate-provider archive plan did not list these repairs; the later single-year plan did record these repairs. These dependency checks reuse saved cases and model roles and were assembled after the earlier analyses, not prospectively registered.

## S18. Annual support and hidden-peak diagnostics

The projection-stage and peak-visibility tables use common finite cases. The peak-visibility stratum depends on the withheld complete reference and is unavailable for an operational decision. Repeated masks can place the same physical station-year in both strata.

The failure-strata table reports case means. The saved summary also contains group-equal means and group-bootstrap intervals, and the leave-one-region-out table repeats the D2-minus-classical comparison after removing one region at a time. These are descriptive summaries of the sampled archive, not independent-climate guarantees. The peak-visibility stratum uses the complete reference and is unavailable at decision time; the gap, overlap and donor bins were chosen after inspection.

**Table S28.** Replay of annual interval stages on common finite-bound cases. Raw is the original model output; stochastic raw quantiles use a finite set of imputations that retain observed days. First applies the first explicit bound projection; Pre-final applies the fixed projection-aware offsets and sorts; Final applies the bound again. New is the count first becoming zero-width at the final projection. Zero widths use $10^{-8}$ °C tolerance. These are repeated mask cases, and no alternative calibration was fitted.

| Family | Setting | Estimator | $n$ | Raw | First | Pre-final | Final | New |
|---|---|---|---|---|---|---|---|---|
| calendar | forward | D2 | 3674 | 0 | 2 | 1096 | 1476 | 380 |
| calendar | forward | Stochastic | 3674 | 3090 | 3090 | 2977 | 2977 | 0 |
| calendar | offline | D2 | 6919 | 0 | 0 | 4402 | 4538 | 136 |
| calendar | offline | Stochastic | 6919 | 5818 | 5818 | 5482 | 5482 | 0 |
| crossed | forward | D2 | 3683 | 0 | 0 | 227 | 323 | 96 |
| crossed | forward | Stochastic | 3683 | 1362 | 1362 | 1091 | 1091 | 0 |
| crossed | offline | D2 | 6919 | 0 | 1 | 1376 | 1391 | 15 |
| crossed | offline | Stochastic | 6919 | 2491 | 2491 | 2030 | 2030 | 0 |

**Table S29.** Retrospective annual performance on the two-estimator common finite-bound cohort, stratified using the complete reference. Match means that the observed lower bound equals the complete reported H30 within $10^{-8}$ °C; excess means the reference exceeds that bound. This outcome-defined stratum is unavailable at deployment. A station-year can occur in both strata under different masks. Intervals in brackets resample connected groups with 2,000 repetitions and are descriptive under the sampled archive, not independent-climate guarantees.

| Family | Setting | Stratum | Method | $n$ | Groups | MAE | Coverage | Group interval |
|---|---|---|---|---|---|---|---|---|
| calendar | forward | Match | D2 | 3125 | 17 | 0.000 | 100.0% | [100.0, 100.0] |
| calendar | forward | Match | Stochastic | 3125 | 17 | 0.000 | 99.9% | [99.6, 100.0] |
| calendar | forward | Excess | D2 | 549 | 15 | 0.191 | 88.3% | [84.6, 93.2] |
| calendar | forward | Excess | Stochastic | 549 | 15 | 0.151 | 87.8% | [84.1, 93.5] |
| calendar | offline | Match | D2 | 5891 | 18 | 0.000 | 99.9% | [99.6, 100.0] |
| calendar | offline | Match | Stochastic | 5891 | 18 | 0.000 | 99.9% | [99.8, 100.0] |
| calendar | offline | Excess | D2 | 1028 | 16 | 0.122 | 90.6% | [86.8, 92.9] |
| calendar | offline | Excess | Stochastic | 1028 | 16 | 0.120 | 83.1% | [79.8, 88.7] |
| crossed | forward | Match | D2 | 1537 | 17 | 0.012 | 99.3% | [98.9, 99.8] |
| crossed | forward | Match | Stochastic | 1537 | 17 | 0.025 | 94.4% | [89.1, 99.3] |
| crossed | forward | Excess | D2 | 2146 | 17 | 0.378 | 83.2% | [72.6, 93.0] |
| crossed | forward | Excess | Stochastic | 2146 | 17 | 0.345 | 81.6% | [72.2, 91.7] |
| crossed | offline | Match | D2 | 2787 | 18 | 0.008 | 99.3% | [98.9, 99.6] |
| crossed | offline | Match | Stochastic | 2787 | 18 | 0.005 | 99.7% | [99.5, 99.9] |
| crossed | offline | Excess | D2 | 4132 | 18 | 0.285 | 90.7% | [88.5, 93.6] |
| crossed | offline | Excess | Stochastic | 4132 | 18 | 0.289 | 87.9% | [84.6, 91.3] |

**Table S30.** Descriptive failure strata for the crossed forward controlled gaps. Peak visibility uses the withheld complete reference. "At least half" includes overlap equal to 0.5. Repeated masks are not independent rivers or climates, and small regional samples do not identify a climate mechanism.

| Stratum | Method | Cases | Stations | Groups | MAE (°C) | 90% coverage | Width (°C) |
|---|---|---|---|---|---|---|---|
| reference peak: reference peak above bound | D2 | 2,146 | 41 | 17 | 0.378 | 83.2% | 1.30 |
| reference peak: reference peak above bound | Classical | 2,146 | 41 | 17 | 0.345 | 81.6% | 1.12 |
| reference peak: reference peak at bound | D2 | 1,537 | 41 | 17 | 0.012 | 99.3% | 0.17 |
| reference peak: reference peak at bound | Classical | 1,537 | 41 | 17 | 0.025 | 94.4% | 0.10 |
| longest gap: 30 d | D2 | 1,234 | 41 | 17 | 0.157 | 92.6% | 0.70 |
| longest gap: 30 d | Classical | 1,234 | 41 | 17 | 0.145 | 88.2% | 0.58 |
| longest gap: 31–60 d | D2 | 1,262 | 41 | 17 | 0.225 | 90.3% | 0.84 |
| longest gap: 31–60 d | Classical | 1,262 | 41 | 17 | 0.213 | 88.5% | 0.71 |
| longest gap: 61–90 d | D2 | 1,187 | 41 | 17 | 0.295 | 86.9% | 0.94 |
| longest gap: 61–90 d | Classical | 1,187 | 41 | 17 | 0.279 | 84.1% | 0.80 |
| training peak overlap: at least half | D2 | 1,874 | 41 | 17 | 0.350 | 87.1% | 1.36 |
| training peak overlap: at least half | Classical | 1,874 | 41 | 17 | 0.332 | 82.2% | 1.17 |
| training peak overlap: no overlap | D2 | 1,232 | 41 | 17 | 0.028 | 97.1% | 0.06 |
| training peak overlap: no overlap | Classical | 1,232 | 41 | 17 | 0.025 | 96.5% | 0.04 |
| training peak overlap: partial | D2 | 577 | 41 | 17 | 0.241 | 83.9% | 0.70 |
| training peak overlap: partial | Classical | 577 | 41 | 17 | 0.218 | 82.1% | 0.55 |

**Table S31.** Descriptive failure strata for the crossed forward controlled gaps. Peak visibility uses the withheld complete reference. "At least half" includes overlap equal to 0.5. Repeated masks are not independent rivers or climates, and small regional samples do not identify a climate mechanism. Continued: visible donors and region.

| Stratum | Method | Cases | Stations | Groups | MAE (°C) | 90% coverage | Width (°C) |
|---|---|---|---|---|---|---|---|
| visible donors: 0 | D2 | 14 | 2 | 2 | 0.270 | 92.9% | 0.94 |
| visible donors: 0 | Classical | 14 | 2 | 2 | 0.327 | 85.7% | 0.82 |
| visible donors: 1–3 | D2 | 1,012 | 17 | 13 | 0.194 | 89.4% | 0.68 |
| visible donors: 1–3 | Classical | 1,012 | 17 | 13 | 0.172 | 89.2% | 0.65 |
| visible donors: 4+ | D2 | 2,657 | 29 | 8 | 0.236 | 90.1% | 0.88 |
| visible donors: 4+ | Classical | 2,657 | 29 | 8 | 0.226 | 86.1% | 0.71 |
| region: Switzerland (FOEN) | D2 | 1,119 | 10 | 2 | 0.100 | 98.7% | 0.77 |
| region: Switzerland (FOEN) | Classical | 1,119 | 10 | 2 | 0.103 | 98.1% | 0.62 |
| region: Bavaria | D2 | 1,983 | 24 | 8 | 0.315 | 82.9% | 0.85 |
| region: Bavaria | Classical | 1,983 | 24 | 8 | 0.279 | 77.9% | 0.69 |
| region: USGS HUC03 | D2 | 92 | 1 | 1 | 0.072 | 100.0% | 0.53 |
| region: USGS HUC03 | Classical | 92 | 1 | 1 | 0.063 | 100.0% | 0.56 |
| region: USGS HUC04 | D2 | 81 | 1 | 1 | 0.075 | 100.0% | 0.98 |
| region: USGS HUC04 | Classical | 81 | 1 | 1 | 0.098 | 100.0% | 0.91 |
| region: USGS HUC10 | D2 | 180 | 2 | 2 | 0.233 | 94.4% | 1.10 |
| region: USGS HUC10 | Classical | 180 | 2 | 2 | 0.359 | 93.3% | 1.09 |
| region: USGS HUC17 | D2 | 228 | 3 | 3 | 0.161 | 97.4% | 0.75 |
| region: USGS HUC17 | Classical | 228 | 3 | 3 | 0.140 | 96.5% | 0.72 |

**Table S32.** Exploratory source-only annual interval expansion on two-method common cases. Supported cases have at least three disjoint source groups and a positive empirical correction; unavailable and zero-correction regions remain in the denominator. Before and after use exactly the same supported cases. The source-group-weighted empirical correction has no finite-sample or climate-transfer guarantee; it was explored after inspecting the principal results. Coverage is percent, width and interval score are in °C.

| Family | Setting | Method | Eligible | Supported | Coverage | Width | Interval score |
|---|---|---|---|---|---|---|---|
| calendar | forward | D2 | 3676 | 0 | – | – | – |
| calendar | forward | Stochastic | 3676 | 0 | – | – | – |
| calendar | offline | D2 | 6926 | 0 | – | – | – |
| calendar | offline | Stochastic | 6926 | 0 | – | – | – |
| crossed | forward | D2 | 3683 | 92 | 100.0→100.0 | 0.532→0.543 | 0.532→0.543 |
| crossed | forward | Stochastic | 3683 | 92 | 100.0→100.0 | 0.555→0.567 | 0.555→0.567 |
| crossed | offline | D2 | 6919 | 0 | – | – | – |
| crossed | offline | Stochastic | 6919 | 0 | – | – | – |

## S19. Source role audit for partitioned forward models

We re-derived the seed-zero bootstrap weights and checked them against the roles saved in 98 model bundles for the four evaluated source splits. The selected held test stations and their recorded connected groups did not enter any source role. However, in the before-2010 split, the role function defines held groups from test-period cases; training-only groups in the same named region may therefore remain in the source. The table reports positive-weight same-region source cases, not held-target observations. This is not a uniformly held-region design; the distinction affects the interpretation of forward transfer. It does not establish leakage of target-station hidden values or physical independence across groups.

## S20. Post-hoc strict-region sensitivity

We refitted the two affected before-2010 seed-zero models after removing every source connected group that appears in the named target region. The original assignment and bootstrap multiplicities of other groups were retained; no replacement groups were drawn. Source interval and release thresholds were recalculated from the filtered roles. In the whole-archive experiment, only the three affected HUC17 stations were reanalysed, on the same 120 original masks; all other original results were retained. Each strict run checked the original mask and the absence of hidden target values. Both estimators were compared on the same station–repetition cases available in both versions. For forward controlled gaps this common cohort comprises 738 cases at 41 stations. Original and strict D2 trend RMSE are 0.1638 and 0.1645; the corresponding stochastic RMSE values are 0.1466 and 0.1466 °C per decade.

The future-year release analysis replaces predictions in both affected regions while preserving all target cases and the other regional models. Across 27,846 forward controlled-

gap cases, the three-output rule publishes 63.1% and 64.5% in the original and strict comparisons, with 10.9% and 11.3% observed failures among the published values. In HUC17, publication changes from 62.7% to 71.6%, and the selected error rate from 6.0% to 8.2%. The selected sets differ, so the conditional rates cannot be interpreted without retention. The strict refits use fewer source observations; these paired comparisons do not identify the causal effect of an individual group and do not establish external validity or safety. They do not recompute every analysis in this paper under strict regional exclusion. Case counts contain repeated masks and are not independent climates.

The source-role comparison additionally checked held test station identities across all source regions, the positive bootstrap weights and the effective split of classifier and probability-calibration groups. Only the two before-2010 models changed positive source weights; zero-weight role membership still matters for downstream group-list use, so this does not prove every downstream result unchanged.

The release-then-trend pipeline was replayed for all 120 HUC17 settings with the original masks, source-specific strict limits and newly fitted model. Original annual states reproduced for 1,440 station–year records. Across the affected records, 39 observed-bound outputs become numerical estimates; the other observed-bound and estimate states remain unchanged. The original and strict pooled controlled-gap comparisons comprise 820 common station–repetition cases and give released-values-plus-bounds working-likelihood trend RMSE 0.2024 and 0.1986 °C per decade, respectively, with mean working interval widths 3.26 and 3.29. These working intervals include regression variation and do not establish calibrated missingness-only coverage.

In the outcome-defined stratum with an unobserved peak, the 2,146 common forward controlled-gap annual cases have D2 nominal 90% coverage 83.2% originally and 82.9% under the stricter source. This retrospective stratum is unavailable to a decision maker and repeated masks are not independent climate realizations. The affected HUC17 source rule-calibration pool contains at most one eligible connected group before model prediction, below the minimum of three for both annual expansion and task calibration; strict refitting cannot change that source-group upper bound, although the corresponding radius and errors were not recalculated.

For forward controlled gaps, the same 818 station–repetition cases from 17 connected groups have every annual, tail and rank-coupling output in both source versions. Under constant tails, nominal 95% D2 trend coverage is 83.0% originally and 83.3% strictly; normal-spacing tails yield 88.8% and 88.9%, with widths 0.451 and 0.454 and interval scores 0.860 and 0.862. In the separate normal-tailed rank experiment, independent ranks yield 88.0% and 87.9% coverage and residual blocks yield 89.0% and 89.0%. Original tail results reproduced on all 480 affected tail rows. These comparisons use the same random ranks within each tail or coupling experiment, but the tail and coupling experiments use different draws; they do not identify additive causes of undercoverage. Descriptive connected-group bootstrap intervals for strict-minus-original metrics accompany the aggregate results.

For the forward controlled-gap D2 versus stochastic comparison, 738 cases at 17 connected groups are available under both source versions and both estimators. The pooled D2-minus-stochastic MSE difference is +0.005350 originally and +0.005559 strictly; the descriptive connected-group bootstrap interval for the strict result is [+0.000957, +0.010431] $(^{\circ}\mathrm{C/decade})^2$. The same resampled groups are used for both source versions. Removing

Bavaria from these fixed predictions leaves 331 cases in 9 groups and changes the difference from -0.000143 originally to +0.000323 strictly. For empirical calendars the corresponding original and strict differences are -0.000213 and -0.000159 on 332 cases. These small changes can alter the sign of the regional comparison; the leave-one-region-out exercise does not refit source models and does not establish independent climates or significance.

The small controlled-gap stress comparison reanalyses 20 settings for one HUC17 station while retaining other stations' original results. Among 30 common baseline/stress station–repetition cases, original and strict baseline D2 RMSE are 0.0794 and 0.0803; for synchronized donor gaps they are 0.1164 and 0.1159, and for heat-associated gaps 0.0885 and 0.0902 °C per decade. Original scenario metrics and masks reproduce. The heat-associated scenario changes both the identity and potentially the number of masked years, so these are exploratory paired scenario comparisons, not isolated causal effects or independent climate tests.

A separate recalculation of the transductive, same-number numerical-output comparator under the strict source model publishes 17,961 controlled-gap values: three-output and score-only failure rates are 11.26% and 7.15%, respectively. The score threshold uses the target feature distribution and the three-output count; counts and selected cases differ between source versions, and lower-bound information is not held equal.

**Table S33.** Seed-zero before-2010 model roles with positive-weight source cases in the named target region. Bootstrap-weighted counts include repeated role draws and do not represent independent samples.

| Region | Role | Groups | Stations | Cases | Weighted cases |
|---|---|---|---|---|---|
| nve_hydapi | interval_calibration | 2 | 3 | 431 | 695 |
| usgs_huc17 | estimator | 1 | 1 | 140 | 280 |

## S21. Source-provider dependence of whole-archive models

The 42 whole-archive target stations belong to providers with recorded reuse paths, but source-model training has additional provenance. Recomputing the source role assignments and checking them against all 13 saved seed-zero model bundles shows that 13 estimator roles and 6 interval-calibration roles have positive-weight LUBW source cases. Source cases are repeated masks; their counts are not independent observations. Positive source weights establish model dependence but do not resolve the legal status of any particular derivative. Individual source, model, target, mask and reviewer-access rights still require confirmation. A post-hoc candidate-provider strict-region refit is reported separately and does not replace the main results. A subsequent post-hoc audit reconstructed donor visibility for all 194,370 eligible candidate source-case rows using the saved masks and processed panels, and checked the counts against the stored cases. In 39 of 52 mode–region–role cells, some positive-weight rows had a finite donor-regression estimate while a visible donor was absent from the original three local catalogs. These are repeated rows, not independent sites or evidence of unauthorized use. The 73 such USGS panel-column occurrences correspond to 69 identifiers, all returned by the USGS site metadata service; one GKD column has a matching project-generated network-QC record. The independent check re-evaluated 156 selected source cases and all 52 role cells. Site-number existence and project QC metadata do not establish the original provenance of cached temperature values, complete processing lineage, or permission to redistribute derived models. The public site metadata snapshot can be replayed separately; source panel mappings remain local. After the metadata audit, we retrieved current USGS daily-mean water temperatures for those 69 sites. Of 293,851 finite local panel-column days, 292,285 had a single distinct finite USGS value and all

agreed within $10^{-6}$ °C. Another 1,525 had multiple distinct USGS values; every local value matched at least one alternative, but that post-hoc match does not determine which series was used historically. Their median within-day range was 0.2 °C; 41 local days had no finite returned value. These are repeated panel-column days, not independent site-days. A separate implementation recomputed all comparisons. The current public responses are included for source-only replay; they do not establish historic data vintage, sensor quality or derivative permissions.

## S22. Candidate-provider, strict-region source sensitivity

After observing the original results and LUBW source dependence, we refitted all 13 whole-archive source models on seven candidate providers with recorded reuse paths. Role assignments and bootstrap draws were recomputed using the filtered data, and every group occurring in the target region was excluded from source roles. An independent role recount found no positive-weight target-region source case or post-2009 forward source case. Original target masks and daily features were reused. All 3,320 archive settings, masks and the unchanged plug-in baselines were checked. Removing providers, changing source assignments and strict regional exclusion jointly change the analysis; the result is not a causal attribution to one of them. The filtered inputs remain processed study data, not provider-native raw records. This after-the-fact sensitivity neither establishes legal redistribution rights nor independent climate transfer, and it does not replace the original analysis. A descriptive connected-group bootstrap resamples the same groups for both source versions and both estimators. It does not establish independent climates or a population transfer guarantee. The corresponding paired MSE differences and bootstrap intervals are supplied with the candidate-source summary.

**Table S34.** LUBW cases with positive source-role weights, not independent station counts. Groups with bootstrap multiplicity zero do not contribute.

| Mode | Region | Estimator case rows | Interval calibration case rows |
|---|---|---|---|
| forward | foen | 383 | 0 |
| forward | gkd_bayern | 287 | 0 |
| forward | usgs_huc03 | 383 | 0 |
| forward | usgs_huc04 | 96 | 0 |
| forward | usgs_huc10 | 191 | 96 |
| forward | usgs_huc17 | 191 | 0 |
| offline | foen | 288 | 311 |
| offline | gkd_bayern | 862 | 2732 |
| offline | syke | 1030 | 311 |
| offline | usgs_huc03 | 3475 | 0 |
| offline | usgs_huc04 | 1461 | 168 |
| offline | usgs_huc10 | 1150 | 0 |
| offline | usgs_huc17 | 1246 | 2732 |

**Table S35.** Trend RMSE in °C per decade on the four-way common cases: both methods and both source-model versions must be available. Repetitions are not independent climates.

| Family | Mode | Cases | Groups | D2 orig. | Class. orig. | D2 cand. | Class. cand. |
|---|---|---|---|---|---|---|---|
| calendar | forward | 701 | 17 | 0.06178 | 0.05134 | 0.06033 | 0.05822 |
| calendar | offline | 835 | 18 | 0.00873 | 0.01124 | 0.00890 | 0.01119 |
| crossed | forward | 738 | 17 | 0.16381 | 0.14658 | 0.15806 | 0.15600 |
| crossed | offline | 838 | 18 | 0.02780 | 0.03208 | 0.02868 | 0.03195 |

**Table S36.** Nominal 95% D2 trend-interval results on cases where both versions are available. Width is in °C per decade; this comparison is descriptive and the selected cases may differ from the four-way point cohort.

| Family | Mode | Cases | Cover orig. | Cover cand. | Width orig. | Width cand. |
|---|---|---|---|---|---|---|
| calendar | forward | 820 | 96.0% | 95.7% | 0.116 | 0.115 |
| calendar | offline | 840 | 95.0% | 95.2% | 0.018 | 0.018 |
| crossed | forward | 820 | 88.0% | 87.6% | 0.450 | 0.408 |
| crossed | offline | 840 | 94.0% | 92.9% | 0.089 | 0.086 |

**Table S37.** Symmetric MSE decomposition on cases with all four retention-by-estimator cells available under both versions. Fractions divide the mean retention contribution by mean total improvement; they are not causal or universal.

| Family | Mode | Estimator | Cases | Groups | Original fraction | Candidate fraction |
|---|---|---|---|---|---|---|
| calendar | forward | D2 | 820 | 17 | 0.799 | 0.783 |
| calendar | forward | classical | 701 | 17 | 0.744 | 0.767 |
| calendar | offline | D2 | 840 | 18 | 0.887 | 0.891 |
| calendar | offline | classical | 835 | 18 | 0.931 | 0.930 |
| crossed | forward | D2 | 807 | 17 | 0.939 | 0.937 |
| crossed | forward | classical | 725 | 17 | 0.925 | 0.931 |
| crossed | offline | D2 | 840 | 18 | 0.930 | 0.933 |
| crossed | offline | classical | 838 | 18 | 0.942 | 0.942 |

## S23. Annual uncertainty under candidate-source refits

The follow-up compares the same target cases where both methods and both source versions have finite seven-quantile distributions and a finite observed bound. The original 24 annual summary cells were independently reproduced on their original cohorts. Target identities, bounds and connected groups agree between versions. The decomposition by whether the complete reference exceeds the visible bound uses the withheld reference and is unavailable at decision time. Candidate-versus-original differences are post hoc; changed source providers, role assignments and strict region exclusion cannot be separated. The quoted CRPS uses the stated normal-spacing tail continuation, and no new calibration or independent climate validation is established. For forward controlled gaps with hidden peaks, the paired candidate-minus-original classical coverage difference is -3.4 percentage points with descriptive connected-group bootstrap interval [-8.6, 2.0]. Removing FOEN or Bavaria from the fixed evaluation cases changes that difference to +0.3 or -8.9 percentage points, respectively. Regions were not refitted in this deletion, and connected groups are not proven independent climates.

## S24. Single-year source expansion for annual intervals

After reviewing prior annual and trend results, we used single-year masked source cases from the candidate models' rule-calibration groups. These groups do not enter the estimators, original interval calibration or risk-learning roles; every target-region group is excluded. Some groups contributed to the separate risk-rule component; the pool is not an unopened independent dataset. Forward source cases are from before 2010. For each family, mode, region and estimator we estimate an equal-group-weighted empirical 90th percentile of nonnegative interval misses. A correction is applied only with at least three

**Table S38.** Annual 90% interval comparison on the same four-way common cases. Hidden peak means that the complete reported H30 exceeds the observed bound; this is an outcome-defined stratum. Widths are in °C. Repeated masks are not independent climates.

| Family / mode | Stratum | Method | Cases | Groups | Cover old | Cover new | Width old | Width new |
|---|---|---|---|---|---|---|---|---|
| calendar / forward | all | D2 | 3674 | 17 | 98.3% | 97.9% | 0.118 | 0.135 |
| calendar / forward | all | classical | 3674 | 17 | 98.1% | 97.9% | 0.088 | 0.093 |
| calendar / forward | hidden | D2 | 549 | 15 | 88.3% | 86.3% | 0.690 | 0.680 |
| calendar / forward | hidden | classical | 549 | 15 | 87.8% | 86.7% | 0.566 | 0.610 |
| calendar / offline | all | D2 | 6919 | 18 | 98.5% | 98.5% | 0.095 | 0.093 |
| calendar / offline | all | classical | 6919 | 18 | 97.4% | 97.7% | 0.066 | 0.069 |
| calendar / offline | hidden | D2 | 1028 | 16 | 90.6% | 90.8% | 0.573 | 0.578 |
| calendar / offline | hidden | classical | 1028 | 16 | 83.1% | 85.1% | 0.427 | 0.434 |
| crossed / forward | all | D2 | 3683 | 17 | 90.0% | 89.2% | 0.826 | 0.734 |
| crossed / forward | all | classical | 3683 | 17 | 87.0% | 87.3% | 0.693 | 0.650 |
| crossed / forward | hidden | D2 | 2146 | 17 | 83.2% | 81.9% | 1.299 | 1.161 |
| crossed / forward | hidden | classical | 2146 | 17 | 81.6% | 78.2% | 1.121 | 1.078 |
| crossed / offline | all | D2 | 6919 | 18 | 94.1% | 92.8% | 0.730 | 0.701 |
| crossed / offline | all | classical | 6919 | 18 | 92.7% | 92.4% | 0.660 | 0.639 |
| crossed / offline | hidden | D2 | 4132 | 18 | 90.7% | 88.6% | 1.171 | 1.127 |
| crossed / offline | hidden | classical | 4132 | 18 | 87.9% | 87.4% | 1.068 | 1.037 |

available source groups and a positive radius. This rule is exploratory, and the source cases hide a single year rather than an entire degraded archive. It neither establishes group exchangeability nor identifies a calibrated joint trend distribution. At most eight forward groups are assigned to this role, below the nine needed for a finite one-maximum-per-group 90% rank; the group-maximum diagnostic was not applied. Re-prediction of the 52 source support cells, verification of individual target intervals and a separate two-method support-intersection check passed. Group-resampling and leave-region-out summaries are supplied; they do not prove independent climates. Unsupported cells and unchanged all-case baselines remain visible. The source group-mass empirical percentile is not a finite-sample coverage guarantee.

## S25. Trend sensitivity to annual interval expansion

After examining the source-only annual correction and the earlier trend results, we propagated it through the original 256-draw normal-spacing quantile representation. The endpoint-only scheme moves the 5th and 95th quantiles while holding the five interior quantiles fixed. The tapered scheme also shifts the 10th, 25th, 75th and 90th quantiles by 0.8, 0.4, 0.4 and 0.8 times the radius, respectively; the median and complete-year values are unchanged. Both schemes use the same adjusted 90% endpoints and respect observed lower bounds. Their interior and normal-spacing outer tails differ. The same independent, auxiliary shared-parameter and residual-block rank matrices are reused within each scheme; no calibrated joint distribution is implied. Among 820 forward controlled-gap station–repetition settings, 760 belong to regions with source support for both methods. Seventy-three of these lack at least one annual distribution, and two have no incomplete year, leaving 685 common nontrivial settings. Original saved trend intervals are reproduced

**Table S39.** Exploratory interval expansion: same target cases must be supported for both methods. Baseline and expanded scores use the identical supported subset; all-case baselines are retained separately in the supplied CSV. Hidden-peak strata depend on the complete reference. Widths are in °C.

| Family / mode | Stratum | Method | Eligible | Supported | Groups | Cover before | Cover after | Width before | Width after |
|---|---|---|---|---|---|---|---|---|---|
| calendar / forward | all | D2 | 3674 | 0 | 0 | – | – | – | – |
| calendar / forward | all | classical | 3674 | 0 | 0 | – | – | – | – |
| calendar / forward | hidden | D2 | 549 | 0 | 0 | – | – | – | – |
| calendar / forward | hidden | classical | 549 | 0 | 0 | – | – | – | – |
| calendar / offline | all | D2 | 6919 | 0 | 0 | – | – | – | – |
| calendar / offline | all | classical | 6919 | 0 | 0 | – | – | – | – |
| calendar / offline | hidden | D2 | 1028 | 0 | 0 | – | – | – | – |
| calendar / offline | hidden | classical | 1028 | 0 | 0 | – | – | – | – |
| crossed / forward | all | D2 | 3683 | 3411 | 14 | 88.7% | 89.9% | 0.705 | 0.755 |
| crossed / forward | all | classical | 3683 | 3411 | 14 | 86.6% | 89.1% | 0.608 | 0.696 |
| crossed / forward | hidden | D2 | 2146 | 1982 | 14 | 81.0% | 82.9% | 1.110 | 1.171 |
| crossed / forward | hidden | classical | 2146 | 1982 | 14 | 76.9% | 81.3% | 1.009 | 1.115 |
| crossed / offline | all | D2 | 6919 | 0 | 0 | – | – | – | – |
| crossed / offline | all | classical | 6919 | 0 | 0 | – | – | – | – |
| crossed / offline | hidden | D2 | 4132 | 0 | 0 | – | – | – | – |
| crossed / offline | hidden | classical | 4132 | 0 | 0 | – | – | – | – |

before adjustment. The target selection and calibration remain retrospective. Connected-group paired bootstrap and fixed-prediction leave-one-region-out summaries are supplied. They are descriptive, use only 14 target groups and do not establish independent climates. Annual 90% endpoint correction does not identify the inner or outer distribution, cure annual undercoverage, guarantee trend coverage, or resolve source rights. The underlying case records and rank arrays remain local pending permission.

**Table S40.** Nominal 95% finite-record trend interval comparison on the same 685 station–repetition cases, 38 stations and 14 connected groups. Width and interval score are in °C per decade; repeated masks are not independent climates.

| Method | Rank construction | Marginal scheme | Coverage | Width | Interval score |
|---|---|---|---|---|---|
| D2 | independent | baseline | 87.9% | 0.382 | 0.742 |
| D2 | independent | endpoints | 89.3% | 0.409 | 0.736 |
| D2 | independent | tapered | 88.9% | 0.403 | 0.735 |
| D2 | shared parameter | baseline | 88.0% | 0.379 | 0.741 |
| D2 | shared parameter | endpoints | 89.2% | 0.405 | 0.730 |
| D2 | shared parameter | tapered | 89.2% | 0.399 | 0.733 |
| D2 | residual blocks | baseline | 88.2% | 0.385 | 0.721 |
| D2 | residual blocks | endpoints | 89.8% | 0.411 | 0.708 |
| D2 | residual blocks | tapered | 89.5% | 0.406 | 0.707 |
| classical | independent | baseline | 83.4% | 0.344 | 0.912 |
| classical | independent | endpoints | 86.9% | 0.387 | 0.856 |
| classical | independent | tapered | 85.5% | 0.376 | 0.866 |
| classical | shared parameter | baseline | 82.2% | 0.342 | 0.922 |
| classical | shared parameter | endpoints | 86.3% | 0.383 | 0.863 |
| classical | shared parameter | tapered | 85.0% | 0.373 | 0.872 |
| classical | residual blocks | baseline | 82.6% | 0.349 | 0.901 |
| classical | residual blocks | endpoints | 86.4% | 0.391 | 0.845 |
| classical | residual blocks | tapered | 85.7% | 0.381 | 0.853 |

## S26. External reference qualification

The Greater Wellington intake returned records for nine selected sites. Under the originally specified quality-code-600 criterion no complete reference hot year was qualified; the provider describes code 200 as unverified, while other returned legacy code meanings remain unresolved. A NEON PRT archive supplied half-hourly records at 24 river or stream sites. With one fixed location code per site, no site supplied four complete years under either 48 quality-passing intervals per UTC day or the 46-of-48 sensitivity. The latter is an observed-interval average rather than a complete physical day; a location code does not establish unchanged hardware.

After the original intake and quality screen, we separately tested an optimistic time-support bound on the same nine cached Greater Wellington records: all quality flags were ignored, while the original fixed-NZST cadence and fully observed day rule were retained and finite values required. Across these records there were 10 potentially complete July–June years, with zero to two per site; no site reached four. This post-hoc feasibility screen is conditional on the cached series and these completeness rules, not a quality assessment, reference qualification, or recovery score. The annual counts were recomputed independently from the same cached XML using a separate day-count implementation; the XML parser and source observations were shared.

An initial 15-station USGS intake had already been processed in the national panel, and ten stations had been scored in an earlier external comparison. The subsequent eight-station roster excluded these and other identified source, evaluation and cached-value contacts; no station passed the four-year reference requirement after the documented quality checks. A further five metadata candidates with prior network screening had no

four years of complete approved daily counts in 2010–2024; counts can include estimated values and do not establish subdaily quality. Five selected Réunion Hub'Eau sites provided records with quality code 1, but no site supplied four complete July–June years under the 24-distinct-valid-hour rule; the provider-day time zone is unresolved. The same intake results do not describe the fraction of eligible stations in any complete network.

At ECCC Red River at Emerson, the USGS alias and an upstream source station place the record on a previously encountered river. The files use the word VALIDATED and, for some rows, a station identifier different from the catalog; the mapping to literal V and numeric quality grades and the daylight-saving convention are unresolved. Under the fixed literal V and grade 1/2 rule, zero complete local-clock years qualified. These exclusions are not an assertion that the rejected observations are erroneous; no numerical recovery score was calculated for this site. Data-source citations appear in the main article.

## S27. Prior Australian reference exposure

An earlier Australian recruitment stage identified 80 observation-supported candidate station-years from 52 stations. The historical intake record states that its designated F2 formal test was closed; this audit does not run that model or calculate new F2 errors. Subsequently, a separate, earlier-method Australian evaluation used calendar-year cases from the same acquired source archive. All 80 of the candidate station-years appear in that case file; 79 had at least one case in the old evaluation's eligible case set, including all 10 of the 10 originally designated F2 test-role station-years. The prior evaluation contained 30,728 eligible masked cases across its wider roster. These are previously evaluated records for a different pipeline, not unopened target station-years for a new independent test.

A later Australian development analysis used July–June rather than calendar-year windows. Counting physical time overlap rather than matching the year labels alone, 74 of the original 80 calendar station-years intersect a saved development window, of which 73 intersect at least one eligible development case. A separate Arrow/Python recount reproduced these counts. Matching station and interval does not prove that masked days are identical, that the F2 model was evaluated, or that data leaked into source-model training. The original native-sampling criterion allowed 20 observed hours per day and does not establish a complete 24-hour physical reference. These previously opened Australian records cannot supply a fresh independent confirmation of the current pipeline.

## S28. Exploratory British Columbia daily-reference comparison

All 23 author-selected published stations were retained for qualification. Four stations each have four complete calendar years in the published daily series. The original study accepted daily means based on at least 20 hourly values after researcher quality checks; hourly counts, original flags and the day convention are not available in this derivative. The source-water cleaning used water–air comparisons. A complete published year is not equivalent to 24 independently quality-checked hours per day. The 23-station author cohort was selected for data availability and these results are unregistered. The fixed global source model and independent source-role recalibration are used; local reconstruction models are refit after holding out the scored year. Each year is masked separately, with 30-, 60- and 90-day controlled gaps. NASA POWER air forcing uses coordinates rounded to 0.01 degree from the source article; there are no same-panel donor stations. The score treats the published daily mean as reference, not as error-free physical truth. Repeated gaps in a station-year and the four stations are dependent; three stations lie in the Fraser River

basin (Nechako, Thompson and Fraser). These are descriptive results, without population uncertainty or a natural-outage validation. Across 372 masked cases, independently

**Table S41.** Controlled-gap annual H30 mean absolute errors in degrees Celsius. Each row has four published-daily reference years; mask counts are not independent observations.

| Station | Masks | Plug-in | D2 | Stochastic |
|---|---|---|---|---|
| 08FF003 | 72 | 0.833 | 0.475 | 0.520 |
| 08JA017 | 72 | 0.241 | 0.243 | 0.203 |
| 08LF051 | 72 | 0.383 | 0.207 | 0.220 |
| 08MF040 | 72 | 0.605 | 0.298 | 0.265 |

recomputed 30-day reference maxima, observed-window bounds, missing counts, and 52 point and interval summary metrics agree with stored results within numerical tolerance. As a limited noninterference probe, one 90-day case per station was regenerated after adding 5 degrees C to hidden target values. All 4 probes retained the same decision-time model features and numerical predictions, while the evaluation reference changed. These probes do not establish noninterference for every case, model path, or experiment. The attribution, candidate roster, year-level qualification, individual station summaries and code are included in the local evidence package. Prior-identifier and geographic proximity checks are best effort and cannot prove independence from every previously opened record.

## S29. Nova Scotia verified workbooks and open-portal comparison

Four river locations were fixed from the provider's historical workbook list before opening their temperature sequences. The workbooks are labelled verified, but their inspected headers do not identify per-hour temperature quality or estimated-value flags; one header states UTC–04:00 and the remaining headers do not establish a time basis. We count

complete reported clock-label years, requiring 24 unique hourly labels with finite numeric cells per calendar day, without claiming these labels correspond to 24 independent physical hours or nonestimated values. The openly licensed portal was accessed only after the workbook completeness screen; it is another release of the same monitoring programme, not an independent sensor. Of 14 station-years with complete labels in both releases, 12 have every numeric value equal within the originally specified $10^{-6}$ °C tolerance. These include at least four such years at 2 stations. Across 262,667 shared unique numeric labels, 12,810 differ at that tolerance; 1,086 exceed 0.5 °C in a post-hoc diagnostic, with a maximum 4.493 °C. These comparisons do not identify which version is correct, and clock-label matching does not resolve physical time. The original reference screen did not score a reconstruction method; a later, explicitly exploratory amendment is reported separately. The portal source is Government of Nova Scotia, dataset bkfi-mjgw, Open Government Licence – Nova Scotia; legacy workbook reuse remains unresolved.

## S30. Nova Scotia reported-clock exploratory masking

The preceding qualification did not establish strict reference quality. After inspecting availability and two-release agreement, a separate unregistered exploratory amendment fixed two sites and eight station-years before any model prediction errors were computed. Each selected year has 24 finite reported hourly labels per day and agrees numerically across the two releases of the same monitoring programme. Time zone, estimated-value status and point-level quality remain unresolved. The reported daily value is the mean of the 24 reported hourly values; it is not guaranteed to represent a physical civil day or an error-free temperature. The fixed global source model and source-only interval calibration were reused. Each year was masked separately with 30-, 60- and 90-day controlled gaps,

using the other three selected years for local history; no empirical-calendar family or whole-archive degradation was evaluated. NASA POWER forcing uses local solar time and coordinates rounded to 0.01 degree. The adjacent-normal tail continuation is an assumption for CRPS. The 144 cases are repeated masks at two sites and eight station-years, not 144 independent watersheds. An independent arithmetic audit recomputed reference H30, bounds, masks and reported scores. In two post-hoc hidden-value probes, adding 5 °C to hidden days changed the true H30 but did not change decision features or predictions when identically serialized controls were compared. Those two probes are not an exhaustive leakage proof.

## S31. Nova Scotia reported-day boundary sensitivity

After inspecting the exploratory results, we tested hypothetical daily bins shifted by $-1$, 0 and $+1$ hour relative to the recorded clock labels. We required all original eight station-years and the adjacent boundary hours to be unique, finite and equal within $10^{-6}$ °C across the two releases; missing or disputed hours were not filled. Daily water inputs and local leave-one-year-out models were rebuilt for each shift, with the original masks, global source model, source calibration and NASA POWER local-solar-time forcing held fixed. All 144 masks are available across all three shifts for both methods; the zero-shift replay agrees with the original outputs. Across the eight reported reference years the largest H30 change is 0.0023 °C, while a post-hoc casewise diagnostic finds a largest D2 point-prediction change of 0.094 °C. This does not identify the true timezone or explain the cause of prediction changes; hourly values, quality and meteorological day alignment remain uncertain. The two sites and repeated masks do not justify a population-level ranking or independent-climate inference.

**Table S42.** Retrospective 2009–2017 reported-label completeness and release concordance. Equal means that every hourly numeric value in the common complete-label year matches to $10^{-6}$ °C at the same reported clock label. Quality, estimated-value status, time basis and climate independence are not established.

| Station | Workbook | Portal | Both | Fully equal |
|---|---|---|---|---|
| NS01DL0009 | 4 | 5 | 4 | 4 |
| NS01ED9111 | 2 | 2 | 1 | 1 |
| NS01EF0002 | 4 | 4 | 3 | 3 |
| NS01FB0001 | 6 | 6 | 6 | 4 |

**Table S43.** Reported-clock exploratory controlled-gap scores on identical available cases. Coverage is percent, and MAE, width and interval score are in °C. These are two sites, not independent climatic replications; source quality and time basis remain unresolved.

| Method | Cases | MAE | 90% coverage | 90% width | Interval score |
|---|---|---|---|---|---|
| D2 | 144 | 0.311 | 88.2 | 1.000 | 1.572 |
| Stochastic | 144 | 0.313 | 93.1 | 1.281 | 1.706 |

**Table S44.** Post-hoc descriptive paired results by gap length and site. Counts are repeated masks; group differences are not inferential tests.

| Axis | Level | Cases | D2 MAE | Stochastic MAE | D2 coverage | Stochastic coverage |
|---|---|---|---|---|---|---|
| Gap days | 30 | 48 | 0.247 | 0.231 | 89.6 | 93.8 |
| Gap days | 60 | 48 | 0.297 | 0.319 | 93.8 | 95.8 |
| Gap days | 90 | 48 | 0.387 | 0.388 | 81.2 | 89.6 |
| Station | NS01DL0009 | 72 | 0.400 | 0.393 | 81.9 | 93.1 |
| Station | NS01FB0001 | 72 | 0.221 | 0.232 | 94.4 | 93.1 |

## S32. Nova Scotia field values and reported hourly ranges

After the portal comparisons and model errors were inspected, we obtained a public field-sample dataset from the same provincial monitoring programme (`https://doi.org/10.25976/sfmj-kf56`). For the same two rivers and 2009–2017 there are 146 water-temperature rows on 73 reported station-dates: each date has two rows with identical metadata and numerical values. We do not know whether they represent duplicate records or separate sampling events. All are labelled preliminary and have no recorded sample time or time zone. Dataset metadata describes co-location with automated probes, but does not establish separate instruments, physical clock alignment or point-level quality.

We compare each field value with the minimum and maximum of the 24 distinct finite portal hours bearing the same reported date, without filling incomplete hours. On 67 station-dates with an available range, values lie inside it on 66. In the previously selected eight model-evaluation station-years, all 33 field dates have a 24-hour reported-label range and 32 lie inside; the largest excess is 0.01 °C. This is not a synchronous comparison: an untimed field value can fall outside the sampled hourly extrema, and an unknown day boundary can change date matching. Pooling the preceding, same and following reported dates gives a much wider, three-day range, not a physical-day uncertainty interval.

After viewing the range result, we also searched all hourly values on each reported date: on 12 of the 33 selected station-dates the field value equals at least one hourly value within $10^{-6}$ °C. Searching 24 values gives many opportunities for accidental agreement and does not show simultaneous sampling or common/separate sensors. A separate CSV/datetime implementation reproduces all 18 range-summary and 18 proximity-summary cells using the same source files. The field records' preliminary status, absent clock, possible equipment

dependence and provider quality procedures remain unresolved. None of these counts upgrades the two sites to strict independent references, establishes physical sensor accuracy, or validates H30 recovery.

## S33. Truckee River reported-clock exploratory evaluation

After the original reference attempts, we accessed the California Department of Water Resources 12-station Truckee water-quality dataset through the public DataONE resolver (`https://doi.org/10.6073/pasta/aefcf302d088aeacc40459091d7e990c`). EML documents provider quality checks and labels the observation clock PST; individual estimated or corrected values are not flagged. Earlier project inventories contain downstream USGS observations from the same watershed. On the named coordinate inventory, the nearest previously known station to each DWR station is more than 10 km away, but coordinate separation does not establish distinct rivers, independent source-model groups or climates. The 12 sites are in one managed watershed.

The station list reports an impossible start after its end for one station; we retained that site for the preplanned timestamp screen instead of interpreting the inconsistency as zero observations. All 12 files have at least four years with a complete grid of unique quarter-hour labels, but this is an upper bound: after requiring all 96 reported water values per day to be finite, only 18 of 171 station-years are complete, and only one station has four such years. The selected years are 2011, 2013, 2014 and 2020 at TRB8. The finite reference consists of each reported-day arithmetic mean of 96 values; it is not a verified nonestimated, physically error-free reference. A second CSV/set implementation recomputed every day and year. The EDI EML declares CC0 while the California discovery catalog lists Creative Commons

Attribution; source attribution is retained and the difference requires reconciliation before permanent release.

After observing availability and numerical reference completeness, but before calculating H30 or target prediction errors, we fixed the one-station cohort and source model. We hold each target year out, use the other three selected years as local history, hide the target gap and use no donor stations. Controlled 30-, 60- and 90-day gaps are crossed with a training-period climatological peak window, with two realizations per cell; there are 72 repeated masks, not 72 independent stations or years. The fixed global source model and source-only calibration are reused. The complete-record reference is subject to sensor and provider-processing uncertainty. No trend is estimated from four years.

After viewing these results, we tested the current NASA POWER hourly UTC and LST products at the same rounded coordinates. For January 1 through December 30 of each reference year, all 34,968 finite LST values equal UTC values eight hours later within $10^{-6}$ °C; 24-hour LST means differ from the originally retrieved daily product by at most 0.005 °C over 1,457 days. Hourly and daily API versions are v2.10.3 and v2.10.1 respectively. If the provider's PST labels denote fixed UTC-8, the two day boundaries agree; the actual water sensor's daylight handling or other clock shifts remain unverified. These numerical checks do not validate either product's physical accuracy or historical version.

Table S46 summarizes the same 72 available cases. D2 coverage is 16/18 in 2011 and 18/18 in each of the other three years; pooling repeated masks does not establish interval calibration or generality. Independent arithmetic reconstructed all 72 references, bounds and score cells. A single post-hoc 90-day hidden-peak perturbation left model inputs and

predictions unchanged while altering the hidden reference; it is a diagnostic, not exhaustive proof against leakage. No group-bootstrap interval is reported for one watershed.

In a further post-hoc stratification, the complete reference equals the observed bound in 35 masks and exceeds it in 37. In the latter, D2 / calibrated-stochastic MAE is 0.246 / 0.356 °C; their 90% coverage is 36/37 / 37/37 and mean width is 2.046 / 2.740 °C. This stratification uses the hidden complete reference and is unavailable as a decision-time rule. All masks still reuse only four years and one station; the selected comparison and apparent coverage cannot establish population or climate-transfer performance.

After the model results were viewed, a Water Quality Portal search identified a field-measurement record at the same DWR station identifier (`https://www.waterqualitydata.us/provider/STORET/CALWR_WQX/CALWR_WQX-G7166310/`). The published coordinates differ by approximately 66 m; the actual sampling positions are unknown. The query returned 60 accepted, actual field-temperature observations for 2011, 2013 and 2014, and no records for 2020. DWR metadata records an EXO2 instrument start on 13 June 2019; these field records therefore do not examine the 2020 instrument period. All carry PST labels and EPA thermometric method 170.1, whose description permits different thermometer types (`https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=30000Q10.TXT`). Matching each to the nearest scheduled 15-minute logger time gives 59/60 differences within 0.5 °C; the maximum time offset is six minutes. A further post-hoc timing stratification finds a mean absolute difference of 0.185 °C and a field-minus-logger mean difference of +0.179 °C among 39 exactly coincident labels. The records are from the same agency and station, and do not identify whether a separate instrument was used. Field visits are sparse and may coincide with servicing; these instantaneous differences cannot be converted directly

into uncertainty in a 30-day H30 maximum. The comparison assumes both PST labels refer to the same fixed clock and is not an independent-climate test.

The local candidate includes the public DWR source files, provider metadata, reconstructed daily values, a reduced WQP field snapshot and the retrieved POWER responses; offline replays recalculate the timestamp, finite-value, field-comparison and forcing checks. The state catalog describes a DWR discrete dataset as public domain, but a record-level crosswalk to this WQP feed was not independently established. The package does not reproduce model scores without the restricted global source-model weights and calibration. A request for access to those materials or a permitted replacement is still needed.

## S34. Boundary checks from additional external archives

These screens bound transport of the main claims. They are not additional primary results. Saved extracts, independent arithmetic replays and the qualification ledger remain with the package.

A 2026 HydroShare reservoir collection (Van Horn et al., `https://www.hydroshare.org/resource/430ee557554948b5811830c1e2ec41aa/`) supplied ten candidate files with 69 station-years. Requiring 96 unique finite quarter-hour values per day in $[-5, 50]\,^{\circ}\mathrm{C}$ yielded 16,153 complete days and no complete calendar year. Bounding unknown slots in that same assumed range produced 50 annual H30 intervals whose median width is 25.18°C; four widths are at most 0.01°C. Completeness screening still discards years that contain a fully observed 30-day window. At Rio Chama below Abiquiu, the 2024 visible window is 19.386°C versus a 2022 conditional H30 of 18.461°C.

An exploratory adaptation scored 306 masks in 17 station-years at four sites that lack four complete training years. On a post-hoc subset of 180 masks with reference width at

most 0.1°C, mean absolute-error bounds are [0.299, 0.325] (plug-in), [0.348, 0.374] (D2) and [0.318, 0.343]°C (stochastic). The full 306-case cohort has mean absolute-error upper bounds exceeding 11°C, so rankings remain weakly identified. All 306 cases lie outside a source-estimator training extremum. A one-dimensional training-range warning flags 42 of 868 source errors (4.8% recall) and does not beat score ranking at matched retention. A Swiss/Bavarian history-degradation diagnostic on 15 stations changes D2 MAE from 0.158°C (full history, count 8) to 0.188°C (degraded, count 0); the count-feature contrast is approximately zero. These are repeated masks, not independent climates, and they do not propose a publication rule.

Eight previously opened USGS daily complete years remain outside the original four-complete-history protocol (`https://api.waterdata.usgs.gov/`). Rio Mora 2017 published daily H30 is 14.1100°C versus 14.1066/14.1055°C from fully supported 15-minute series under fixed-standard/civil time. On 54 hourly-supported masks at that station, D2 MAE is 0.180°C against published daily H30; nominal 90% coverage is 52/54 against the daily target and 33/54 against the conditional continuous target in $[-5, 50]\,^{\circ}\mathrm{C}$. Removing nonempty daily qualifiers from histories changes those MAEs from 0.266/0.180/0.174 to 0.242/0.226/0.218°C. Softening the published support by 0.05°C before source recalibration yields 37/54 D2 continuous-target coverage. Daily-target accuracy does not establish equivalent physical-reference accuracy, and none of these checks is independent-climate validation.

**Table S45.** Paired exploratory reported-day boundary sensitivity with identical original masks. Coverage is percent and error, width and interval score are in °C. Local stages are refit; source model, source calibration and forcing are held fixed. Two sites and eight station-years are not independent climatic replications.

| Shift (h) | Method | Cases | MAE | 90% coverage | Width | Interval score |
|---|---|---|---|---|---|---|
| -1 | D2 | 144 | 0.310 | 88.9 | 1.000 | 1.555 |
| -1 | Stochastic | 144 | 0.313 | 92.4 | 1.283 | 1.704 |
| 0 | D2 | 144 | 0.311 | 88.2 | 1.000 | 1.572 |
| 0 | Stochastic | 144 | 0.313 | 93.1 | 1.281 | 1.706 |
| 1 | D2 | 144 | 0.315 | 86.8 | 0.998 | 1.572 |
| 1 | Stochastic | 144 | 0.313 | 93.1 | 1.278 | 1.731 |

**Table S46.** Truckee exploratory annual H30 scores on 72 controlled masks from one station and four years. MAE, bias, width and interval score are °C; coverage is percent for the nominal 90% interval. The plug-in method supplies no interval.

| Method | Available | MAE | Bias | Coverage | Width | Score |
|---|---|---|---|---|---|---|
| Plug-in | 72 | 0.458 | -0.458 | – | – | – |
| D2 | 72 | 0.131 | -0.047 | 97.2 | 1.106 | 1.133 |
| Calibrated stochastic | 72 | 0.188 | -0.100 | 98.6 | 1.459 | 1.464 |

## S35. Exploratory process and numerical diagnostics

These checks were developed after inspecting earlier regional and time-support results. They are exploratory and reuse the opened archive; selection of six stations before a numerical replay is not preregistration of the research questions.

### S35.1. Common support and common-weight standardization

Common geometric cells contain gap length, training-peak overlap, year and visible-donor bins. Screening to shared cells preserves region-specific cell frequencies and is not balancing. The primary standardized target assigns identical weights proportional to the smaller regional cell count; equal-cell weighting is a sensitivity. Joint-cell frequency TV becomes zero under either common target, but within-bin continuous quantities and training histories remain imbalanced. In the geometry min-count analysis, mean training complete-year counts are 9.78 in FOEN versus 22.96 in Bavaria (SMD -3.052); after adding reconstructor they are 8.50 versus 20.18 (SMD -2.707). Saved CSV files also report continuous means, weighted KS distances and missing-weight fractions. A coverage contrast remains under these common geometric-cell weights; residual within-cell and history differences remain, and the comparison does not identify a reconstructor effect or exclude river processes. The absence of a Swiss air-ridge stratum prevents corresponding cross-region comparison of the main Bavarian failure layer.

### S35.2. Annual water–air associations

Pooled annual associations mix station differences and within-station variation. We therefore center both H30 quantities within station and period, before/from 2010, and pool those centered observations. Both periods of a sampled connected group remain paired in

**Table S47.** Descriptive regional coverage standardization on common cells. Min-count weights are the primary common target; equal-cell weights are a sensitivity analysis. Coverage is in percent and differences are percentage points. Geometry cells contain gap, training-peak overlap, year and visible-donor bins. Adding the reconstructor changes the supported population; it does not identify a reconstructor effect.

| Cells | Weights | FOEN | Bavaria | Difference |
|---|---|---|---|---|
| Geometry | Original | 98.75 | 82.78 | 15.97 |
| Geometry | Min-count | 98.72 | 82.68 | 16.04 |
| Geometry | Equal-cell | 99.29 | 81.68 | 17.62 |
| + reconstructor | Original | 99.58 | 94.12 | 5.46 |
| + reconstructor | Min-count | 99.72 | 96.04 | 3.69 |
| + reconstructor | Equal-cell | 99.75 | 93.41 | 6.33 |

Reconstructor common support retains 472/1,119 FOEN and 289/1,983 Bavaria cases. Switzerland has no air-ridge cases, so Bavaria's air-reconstruction failure stratum has no cross-region counterpart.

10,000 bootstrap replicates (seed 20261029); station-paired resampling is an additional sensitivity, not an assumption-free replacement. FOEN has two groups, with only three distinct bootstrap group compositions; its intervals are particularly coarse. Similar annual within-slope point estimates neither establish equivalence nor validate gap-day residuals, heat-peak timing, tails or autocorrelation. Each variable is its own annual maximum and their maximizing dates need not coincide.

**Table S48.** Remaining balance after regional standardization. TV denotes total variation of discrete distributions. History denotes training complete-year count. SMD denotes the FOEN-minus-Bavaria mean difference divided by the pooled population standard deviation in the original two regions, kept fixed across weighting schemes. Full continuous means, missing-weight fractions and weighted Kolmogorov distances are saved with the analysis. Zero joint-cell TV does not ensure balance within bins.

| Cells | Weights | Joint TV | Policy TV | History-bin TV | History SMD |
|---|---|---|---|---|---|
| Geometry | Full | 0.232 | 0.854 | 0.948 | -3.006 |
| Geometry | Common raw | 0.228 | 0.853 | 0.948 | -3.000 |
| Geometry | Min-count | 0.000 | 0.919 | 0.979 | -3.052 |
| Geometry | Equal-cell | 0.000 | 0.780 | 0.916 | -3.176 |
| + reconstructor | Full | 0.884 | 0.854 | 0.948 | -3.006 |
| + reconstructor | Common raw | 0.614 | 0.000 | 0.696 | -2.292 |
| + reconstructor | Min-count | 0.000 | 0.000 | 0.722 | -2.707 |
| + reconstructor | Equal-cell | 0.000 | 0.000 | 0.760 | -2.891 |

### S35.3. Year-set comparisons and retained baselines

The original single random selection and incomplete-year-only spacing rule are retained as labeled legacy comparators in the table below. The exact subset expectation and the two selectors that incorporate fixed complete years are the current exploratory extensions. Paired intervals in this table use seed 2026102969. The following subsection reports the same one-draw contrasts under seed 20261029, including the 468-archive choice and the

**Table S49.** Hidden-reference-peak recovery by region and daily reconstructor. The hidden-peak label uses withheld complete-year H30. Coverage is nominal 90%. Above-interval is the share of complete references hotter than the 95th percentile.

| Region | Reconstructor | Method | Cases | MAE | Bias | 90% coverage | Above interval |
|---|---|---|---|---|---|---|---|
| Switzerland | donor_ridge | D2 | 633 | 0.175 | -0.130 | 98.1% | 0.9% |
| Switzerland | donor_ridge | classical | 633 | 0.180 | -0.133 | 96.7% | 2.7% |
| Bavaria | air_ridge | D2 | 995 | 0.574 | -0.479 | 67.9% | 30.2% |
| Bavaria | air_ridge | classical | 995 | 0.486 | -0.296 | 66.5% | 27.1% |
| Bavaria | donor_ridge | D2 | 163 | 0.249 | -0.129 | 90.8% | 8.6% |
| Bavaria | donor_ridge | classical | 163 | 0.205 | -0.079 | 87.7% | 8.6% |

incomplete-year-only span. The two seeds agree on whether each legacy interval contains zero.

### S35.4. Common gap metric and archive count audit

The primary comparison uses 2010–2025 to match the forward experiment. The observed eligible archive covers 41 forward-cohort stations; only 29 have incomplete years in this period. There are 109 unique incomplete station-years, without artificial mask repetition. The empirical-calendar and controlled-gap summaries contain 3,806 and 3,777 incomplete station-year–replicate rows, respectively, across the same 41-station forward cohort. Their repeated appearances preserve experimental weighting and do not create new independent observations. All three summaries use each year's longest contiguous missing run, including unions of adjacent or overlapping missing spans.

**Table S50.** Annual water–air H30 associations in unique complete station-years. Years and slopes are shown before/from 2010; S/G counts stations/connected groups. Within estimates center both variables within station and period. Slopes and correlations pool the centered annual observations, rather than averaging station coefficients. The paired 95% percentile intervals resample the same connected groups in both periods; only two FOEN groups and eight Bavaria groups are available, so these intervals are descriptive and coarse. Station-resampling sensitivity results are saved separately.

| Region | Fit | Years | S/G | Slopes | $\Delta b$ [95% interval] |
|---|---|---|---|---|---|
| FOEN | Pooled | 98/159 | 10/2 | 1.143/1.176 | +0.033 [-0.005, +0.047] |
| FOEN | Within | 98/159 | 10/2 | 0.715/0.701 | -0.014 [-0.015, -0.005] |
| Bavaria | Pooled | 554/295 | 24/8 | 1.318/1.409 | +0.091 [-0.151, +0.346] |
| Bavaria | Within | 554/295 | 24/8 | 0.814/0.809 | -0.005 [-0.224, +0.149] |

The air and water H30 maxima need not refer to the same dates. Similar annual within slopes do not establish stable daily residuals, hidden-peak timing, tails or autocorrelation, and are not an equivalence test.

As an additional historical description, the observed archive over 1981–2025 contains 220 incomplete station-years at 31 stations; the median annual longest run is 3 days and 11.36% have a run of at least 30 days. This historical period is supplementary rather than the primary comparison to forward masks. The calendar masks are closer to common observed gap conditions on this metric, whereas 30–90-day controlled masks probe the long-gap tail; differing sampling and repeat weights preclude a same-distribution claim.

The shared-year fraction counts station-years in which at least one other station in the same connected group also has an incomplete year (within the same replicate for trial

**Table S51.** Annual water–air correlations and paired changes. Intervals use connected-group resampling, as in Table S50. Correlations describe annual H30 associations at the stated aggregation scale.

| Region | Fit | $r_{<2010}$ | $r_{\geq 2010}$ | $\Delta r$ [95% interval] |
|---|---|---|---|---|
| FOEN | Pooled | 0.842 | 0.817 | -0.025 [-0.129, -0.014] |
| FOEN | Within | 0.855 | 0.725 | -0.130 [-0.130, -0.129] |
| Bavaria | Pooled | 0.649 | 0.708 | +0.059 [-0.027, +0.100] |
| Bavaria | Within | 0.826 | 0.800 | -0.025 [-0.066, +0.051] |

masks). It does not require overlapping missing days and therefore does not establish synchronous daily outages or unavailable donors on the same dates.

Complete-year counts in the cohort comparison count unique complete station-years in the leave-one-year-out source cases, including the held-out year. The previous 30/7 medians counted training years: every one of the 581 stations differs by exactly one year. Correct archive medians are 31/8. Visible-donor summaries derive from leave-one-year-out cases and describe that design, rather than the forward controlled-gap experiment.

### S35.5. Paired time-support diagnostics

These exploratory comparisons use one reproducible random subset per station–replicate archive, shared by the retained analysis. They do not average over all random year selections. The time-balanced heuristic spreads only the added incomplete years; it does

**Table S52.** Paired resampling sensitivity for within-station annual water–air H30 changes (from 2010 minus before 2010). Each resample carries both periods of a selected group or station. Connected groups are the primary sampling unit; station resampling assumes independence across stations. Neither percentile interval establishes equivalence, and FOEN group intervals have only three distinct possible compositions.

| Region | Unit | $n$ | $\Delta b$ [95% interval] | $\Delta r$ [95% interval] |
|---|---|---|---|---|
| FOEN | Group | 2 | -0.014 [-0.015, -0.005] | -0.130 [-0.130, -0.129] |
| FOEN | Station | 10 | -0.014 [-0.043, +0.032] | -0.130 [-0.163, -0.097] |
| Bavaria | Group | 8 | -0.005 [-0.224, +0.149] | -0.025 [-0.066, +0.051] |
| Bavaria | Station | 24 | -0.005 [-0.123, +0.109] | -0.025 [-0.069, +0.022] |

not optimize their placement jointly with the fixed complete years. Its mean span changes only from 13.265 to 13.343 years. Thus year count and time position or slope leverage remain inseparable.

The following 95% percentile intervals resample the 17 connected groups 10,000 times with seed 20261029, carrying every paired station–replicate archive with its group. Each statistic weights archives equally, including repeated masks. These intervals describe the present cohort and do not establish equivalence.

The same-count rules are identical by construction in 337/805 archives (41.9%): 332 retain no incomplete year and 5 retain every incomplete year. Only 468 archives offer a choice between different same-sized subsets. Their estimated RMSEs are 0.3944 (Warm-

**Table S53.** Exploratory finite-record slope comparisons on 805 actual-mask archives in 17 connected groups. Errors are °C per decade. CDF and Sxx optimize the entire retained set including fixed complete years. Uniform exact averages squared errors over feasible same-count sets within each archive, then averages archives and takes the square root. All-year truth-set error is zero by definition. The last column gives comparator-minus-warm-season estimated RMSE with a paired 95% group-percentile interval, not an equivalence test.

| Rule | Years | Estimate | Year-set | Reconst. | Δ RMSE [95% interval] |
|---|---|---|---|---|---|
| Warm-season | 9.59 | 0.4459 | 0.4458 | 0.0023 | Reference |
| Legacy random | 9.59 | 0.4540 | 0.4569 | 0.1030 | 0.0081 [-0.0103, 0.0177] |
| Legacy spacing | 9.59 | 0.4566 | 0.4539 | 0.0873 | 0.0108 [-0.0063, 0.0404] |
| Uniform exact | 9.59 | 0.4492 | 0.4483 | 0.0978 | 0.0033 [-0.0052, 0.0171] |
| CDF | 9.59 | 0.4284 | 0.4199 | 0.1079 | -0.0175 [-0.0274, -0.0049] |
| Sxx | 9.59 | 0.4023 | 0.3931 | 0.1185 | -0.0436 [-0.0645, -0.0173] |
| All years | 13.32 | 0.1697 | 0.0000 | 0.1697 | -0.2762 [-0.3557, -0.1841] |

season 90%), 0.4100 (Same-count random), 0.4150 (Same-count time-balanced) °C per decade. Similar overall point estimates therefore partly reflect identical selections.

For each archive, $S$ is the complete-year slope on the selected set minus the full finite-record slope, $R$ is the estimated slope minus the complete-year slope on that selected set, and $E = S + R$. Consequently,

$$\mathrm{MSE}(E) = \mathrm{MSE}(S) + \mathrm{MSE}(R) + 2\,\mathrm{mean}(SR).$$

**Table S54.** Annual longest contiguous missing run. The first three rows are the 2010–2025 comparison. Observations weight incomplete station-years equally; trial rows weight incomplete station-year–replicates equally. The 1981–2025 row is the longer observed archive on the same statistic. Sampling and repeat weights differ, so this is not a same-distribution test.

| Source | Rows | Stations | Median (d) | 90th percentile (d) | $\geq 30$ d |
|---|---|---|---|---|---|
| Observed archive | 109 | 29 | 4 | 33.6 | 11.93% |
| Empirical calendar | 3,806 | 41 | 6 | 30.0 | 10.25% |
| Controlled gaps | 3,777 | 41 | 60 | 90.0 | 100.00% |
| Observed 1981–2025 | 220 | 31 | 3 | 32.4 | 11.36% |

**Table S55.** Long-complete evaluation stations versus the remainder of the 581-station source archive. Complete-year counts are unique complete station-years, including the held-out year, rather than training-year counts. USGS counts follow named hydrologic-unit regions.

| Panel | Stations | Groups | Median years | Bavaria | Switzerland | USGS |
|---|---|---|---|---|---|---|
| Trend cohort | 42 | 18 | 31 | 24 | 10 | 7 |
| Source remainder | 539 | 183 | 8 | 29 | 1 | 312 |

For the same-count random control, the numerical identity is $0.206128 = 0.208715 + 0.010617 - 0.013204$. The negative cross term shows error cancellation; RMSE differences and normalized squared terms are not independent causal contributions.

**Relationship to the 807-archive core comparison.:** Two of the original common archives have no masked year and empty annual-prediction schemas: Bavaria station 10081004, replicate 6 (14 complete years), and station 11405000, replicate 14 (7 complete years). Their design masks are empty and saved warm-season and all-year slope errors are

**Table S56.** Forward-mask geometry on the 41-station trend cohort with twenty replicates. Year selection after a seven-year burn-in is shared. Segment length and season come from the copied empirical calendars or from the controlled-gap design. Shared-year fraction is the share of masked station-years that are also masked at another station in the same connected group.

| Family | Median length (d) | JJA share | Adjacent years | Shared years | Stations |
|---|---|---|---|---|---|
| Empirical calendar | 4 | 19.9% | 43.9% | 60.9% | 41 |
| Controlled 30–90 d | 60 | 58.5% | 43.5% | 60.1% | 41 |

both zero. The prior diagnostic skipped them because the empty prediction files contain neither a year nor a D2 column. The present 805-archive diagnostic explicitly retains that actual-mask scope; this is not a reconstruction failure. A saved 807-archive sensitivity restores their zero errors to every scheme. The remaining 13 of the 820 forward archives lack an available warm-season comparison and are outside both common denominators.

### S35.6. Earlier uncalibrated-only trajectory diagnostic

The original three-seed check reports raw, uncalibrated endpoints in the fixed 27-case sample. Its small pooled median does not establish stratified stability. On seed 0, six of nine Bavarian upper endpoints move by more than 0.05°C, and the mean 90% width increases from 0.3179 to 0.3686°C. The subsequent experiment section additionally regenerates the original source-calibration roles and re-estimates offsets at each budget; the table below is retained specifically as the uncalibrated reference.

**Table S57.** Bavarian whole-archive stations. Regulation labels are project metadata.

| Code | Group | Latitude | Longitude | Years | Regulation |
|---|---|---|---|---|---|
| 10026293 | g0032 | 48.389 | 9.987 | 1994–2024 | unmatched_gages |
| 10039802 | g0032 | 48.711 | 10.802 | 1997–2025 | unmatched_gages |
| 10046105 | g0032 | 48.757 | 11.425 | 1981–2025 | unmatched_gages |
| 10056302 | g0032 | 48.947 | 12.015 | 1993–2025 | unmatched_gages |
| 10074009 | g0032 | 48.886 | 12.574 | 1981–2024 | unmatched_gages |
| 10081004 | g0032 | 48.826 | 12.961 | 1981–2024 | unmatched_gages |
| 10089006 | g0032 | 48.635 | 13.187 | 1981–2025 | unmatched_gages |
| 10091008 | g0032 | 48.576 | 13.459 | 1981–2025 | unmatched_gages |
| 10098007 | g0032 | 48.514 | 13.727 | 1981–2023 | unmatched_gages |
| 11401009 | g0034 | 47.524 | 10.260 | 1981–2025 | unmatched_gages |
| 11402001 | g0034 | 47.730 | 10.317 | 2001–2025 | unmatched_gages |
| 11405000 | g0034 | 48.371 | 9.988 | 1981–2025 | unmatched_gages |
| 16003003 | g0036 | 47.684 | 11.572 | 1981–2025 | unmatched_gages |
| 16005701 | g0036 | 48.145 | 11.597 | 1981–2024 | unmatched_gages |
| 16007004 | g0036 | 48.528 | 12.133 | 1981–2025 | unmatched_gages |
| 17207508 | g0039 | 48.611 | 13.145 | 1981–2025 | unmatched_gages |
| 18001508 | g0035 | 47.855 | 12.143 | 1981–2020 | unmatched_gages |
| 18008008 | g0035 | 48.562 | 13.444 | 1981–2024 | unmatched_gages |
| 18403002 | g0031 | 47.934 | 12.481 | 1981–2024 | unmatched_gages |
| 24010004 | g0037 | 49.951 | 10.874 | 1981–2025 | unmatched_gages |
| 24022003 | g0037 | 50.031 | 10.221 | 1981–2025 | unmatched_gages |
| 24042000 | g0037 | 49.796 | 9.926 | 1981–2025 | unmatched_gages |
| 24064003 | g0037 | 49.716 | 9.225 | 1981–2025 | unmatched_gages |
| 24405002 | g0033 | 50.197 | 10.075 | 1981–2025 | unmatched_gages |

**Table S58.** Non-Bavarian whole-archive stations, including Switzerland (FOEN) and USGS HUC17. Regulation labels are project metadata.

| Code | Group | Latitude | Longitude | Years | Regulation |
|---|---|---|---|---|---|
| 2016 | g0029 | 47.482 | 8.194 | 2000–2025 | unmatched_gages |
| 2019 | g0029 | 46.746 | 8.092 | 2000–2025 | unmatched_gages |
| 2029 | g0029 | 47.122 | 7.283 | 2000–2025 | unmatched_gages |
| 2030 | g0029 | 46.765 | 7.612 | 2000–2025 | unmatched_gages |
| 2085 | g0029 | 47.056 | 7.184 | 2000–2025 | unmatched_gages |
| 2113 | g0029 | 47.595 | 8.225 | 2000–2025 | unmatched_gages |
| 2126 | g0030 | 47.496 | 8.953 | 2002–2025 | unmatched_gages |
| 2135 | g0029 | 46.933 | 7.448 | 2000–2025 | unmatched_gages |
| 2205 | g0029 | 47.516 | 8.235 | 2000–2025 | unmatched_gages |
| 2457 | g0029 | 46.694 | 7.880 | 2000–2025 | unmatched_gages |
| 1670 | g0087 | 62.598 | 25.739 | 1935–1994 | unmatched_gages |
| 02172020 | g0181 | 33.094 | -79.949 | 1988–2025 | unknown |
| 04137030 | g0166 | 44.473 | -83.571 | 1999–2024 | unknown |
| 06036905 | g0053 | 44.620 | -110.864 | 2003–2025 | regulated |
| 06054500 | g0143 | 46.147 | -111.420 | 2001–2025 | regulated |
| 13342500 | g0054 | 46.448 | -116.828 | 2001–2024 | regulated |
| 14166000 | g0057 | 44.270 | -123.174 | 2003–2025 | regulated |
| 14211500 | g0059 | 45.477 | -122.508 | 2000–2021 | regulated |

**Table S59.** Paired estimated-slope RMSE contrasts, comparator minus warm-season screen (°C per decade).

| Comparator | Difference | 95% group-resampling interval |
|---|---|---|
| Same-count random | +0.0081 | [-0.0106, +0.0176] |
| Same-count time-balanced | +0.0108 | [-0.0060, +0.0394] |
| All supported years | -0.2762 | [-0.3572, -0.1832] |

## S36. Additional exploratory experiments

### S36.1. Outcome-blind year-set selection

All feasible same-count subsets are enumerated while retaining every fixed complete year. There are 5758 subsets across 807 common archives; the 805 actual-mask primary archives contribute 5756 subsets. The two zero-mask settings contribute one subset each. Only 468 actual-mask archives offer a nontrivial choice; the remaining 337 have identical same-count selections by construction. CDF minimizes the retained-versus-full empirical-year-CDF squared distance, including complete years; Sxx maximizes the centered decade-year sum of squares of the entire retained set. Ties use the alternate geometry score and then the earliest incomplete-year tuple. No temperature or error enters either selector. They change 367 and 376 selected sets relative to warm-season screening, respectively. These post-review exploratory rules are not deployment recommendations and do not separately identify effects of year count and temporal position.

Uniform exact means $\sqrt{\mathrm{mean}_a\, E_{U|a}(E_a^2)}$, with an equal-probability set $U$ inside each archive $a$. It is not the mean RMSE over random selections. Over 200 random seeds, the mean RMSE is 0.4505, the root of mean sampled MSE is 0.4506, and the exact conditional root-expected-MSE is 0.4492 °C per decade. The across-seed 2.5th–97.5th

percentiles [0.4359, 0.4671] describe selection variation, not a confidence interval; 62/200 seeds improve on the warm-season point result. Paired intervals in Table S53 instead resample the recorded connected groups 10,000 times, preserving paired archives and their conditional subset-average MSEs before taking roots.

For each fixed set $E = S + R$, hence $\mathrm{MSE}(E) = \mathrm{MSE}(S) + \mathrm{MSE}(R) + 2\,\mathrm{mean}(SR)$. Averaging over uniform subsets preserves this identity; negative cross terms can cancel squared errors. Neither RMSE subtraction nor normalized squared terms are independent causal contributions.

### S36.2. Larger-budget source-role calibration

The source calibration union contains 2370 cases at 19 stations and 102 unique station-years. The Swiss/Bavarian target roles contain 4/4 positive-weight interval-calibration groups under the original saved role assignment. All source calibration years precede 2010; held target groups remain excluded and original integer bootstrap multiplicities are unchanged. For each seed and budget the order is seven raw source quantiles → observed-bound projection → weighted source residual order statistics; target raw quantiles → bound → add source offset → sort → bound. A cell requires at least 50 finite positive-weight cases before multiplicity; otherwise the finite positive calibration pool is used. No target truth is used for offset estimation.

Native20 calls the original bridge with 20 draws, whereas nested20 is the prefix of a 500-draw call; they need not be identical because the random arrays are generated by day. Native20 source and target references are checked separately against original case/repair and archive files. Seeds 0, 1 and 2 share fixed daily models, residual histories and masks. The uncalibrated, fixed-old-offset and updated-source-offset layers score the same target

raw draws. D2 retains its original seven annual quantiles, with no refit or recalibration. Every CRPS uses the common bounded normal-spacing 256-point annual approximation; these are annual scores, not newly computed full-archive trends. The 27 target years span only two connected groups; 500 paths do not prove convergence or broad ranking robustness.

**CRPS integration precision.:** Keeping every saved annual quantile and the tail assumption fixed, we score 783 method–case distributions on 256, 1,024 and 4,096 probability points. An independent order-statistic formula reproduces the original 256-point scores to floating-point precision. The maximum casewise 256-to-4,096 change is 0.000063°C. The fixed-seed-0 updated-source nested20/nested500 comparison crosses the fixed-D2 CRPS at all three grids. This checks integration of the stipulated seven-quantile approximation; it does not validate its tail assumption or establish infinite-trajectory convergence.

### S36.3. Regional and annual water–air influence diagnostics

Removing a connected group and recomputing common cells changes both support and the standardized target population; it is an influence check, not a fixed-population causal contrast. Geometry min-count coverage gaps stay positive, whereas the smaller geometry-plus-reconstructor subset can reverse sign. Training-history imbalance and the missing Swiss air-ridge counterpart remain. Annual water–air sensitivity also depends on whether years or stations receive equal total weight. A group deletion can leave only one Swiss station, and the Bavarian aggregate change masks group influence. None of these checks establishes daily reconstruction stability or equivalence.

### S36.4. Same-day missingness with an evaluable-peer denominator

Peer-evaluable rows require another station in the same connected group with an eligible recorded missingness pattern in that year. Observed peers use the eligible-year roster, which includes incomplete years; trial peers use the complete-reference-year roster on which masks are imposed. Group-years with no such peer are excluded and counted. Same-day overlap means at least one missing day coincides with a peer missing day. Shared station-days count each focal station's missing days that coincide with any peer, divided by its eligible missing station-days. Observed rows are incomplete station-years; trial rows are station-year–replicates. Same-day co-missingness does not identify an outage cause, the donor actually used, or a naturally unobserved failure process.

### S36.5. Empirical-calendar extension and selection opportunity

The calendar extension applies the same three outcome-blind rules to the saved forward empirical-calendar predictions. It does not refit a daily or annual model, change the source calibration or alter the fixed complete years. There are 0.1120 °C per decade of year-set error under warm-season screening and 0.1136 total estimated-slope RMSE on the 818 actual-mask archives. The CDF rule gives 0.0984, the $S_{xx}$ rule gives 0.0859, and retaining every supported year gives 0.0674. Uniform exact averaging gives 0.1355. The corresponding comparator-minus-warm-season paired intervals are -0.0152 [-0.0451, 0.0004] for CDF, -0.0277 [-0.0497, -0.0169] for $S_{xx}$, and -0.0462 [-0.0706, -0.0295] for all years. The CDF interval includes zero; the $S_{xx}$ and all-year intervals lie below zero in this cohort.

The opportunity to select years differs from the control-gap family. Only 280 of the 818 empirical-calendar archives offer a choice between different same-count subsets; 530 already retain every incomplete year and 8 retain none. The extension therefore measures

a narrower selection margin, and it is not an external accuracy validation on genuinely unknown gaps. The control-gap and empirical-calendar cohorts are also not exchangeable: their rows are separate, and subtracting their differences would not identify a causal effect of a missingness mechanism. Within the calendar family, sign disagreement with the complete-reference slope is 3/818 under warm-season screening, 5/818 under CDF, 4/818 under $S_{xx}$ and 0/818 when all supported years are retained. These counts are small and describe this cohort only. They show that aggregate RMSE and direction agreement are different criteria, not that any rule is a safe climate-direction classifier.

### S36.6. Algebraic year-set error and an iid working benchmark

For a fixed station, let $F = Y_i$ be the complete reference-year set and let $S$ be a retained subset that contains every year of $F$ outside the masked years. Write the annual H30 vector as $h$, with entries ordered by year. For each set $A \in \{F, S\}$, define its own centering and slope weights in decade units:

$$x_{A,y} = \frac{y - \bar{y}_A}{10}, \qquad S_{xx,A} = \sum_{z \in A} x_{A,z}^2, \qquad w_{A,y} = \begin{cases} x_{A,y}/S_{xx,A}, & y \in A, \\ 0, & y \notin A. \end{cases} \tag{1}$$

The weights are centered, so $\sum_y w_{A,y} = 0$ and $\sum_y w_{A,y} x_{A,y} = 1$; the intercept is not part of the slope weight vector. The finite-record target is $\beta_F = w_F^\top h$ and the selected-set slope is $\beta_S = w_S^\top h$. Their difference is

$$S_{\text{set}} = (w_S - w_F)^\top h = (w_S - w_F)^\top e, \tag{2}$$

where $e = h - \bar{h}_F \mathbb{1} - \hat{b}_F x_F$ is the residual from the complete-reference linear fit. The second equality is exact because the weight difference sums to zero and is orthogonal to the complete-reference year vector $x_F$: $w_F^\top x_F = 1$ and $w_S^\top x_F = 1$. Thus a constant shift and a strictly linear trend cancel: if the annual sequence were exactly linear and recovery

introduced no error, every retained subset would give the same slope. The observed year-set error is the inner product of the weight difference with the residual, not with the fitted line.

Under the additional working model $h = \alpha\mathbb{1} + bx_F + \varepsilon$ with $\mathrm{Var}(\varepsilon) = \sigma^2 I$ and a year set fixed independently of $\varepsilon$,

$$\mathrm{Var}(S_{\mathrm{set}}) = \sigma^2 \left( \frac{1}{S_{xx,S}} - \frac{1}{S_{xx,F}} \right), \qquad S_{xx,A} = \sum_{y \in A} x_{A,y}^2. \tag{3}$$

This follows from (2) and the exact weight-distance identity $d^\top d = 1/S_{xx,S} - 1/S_{xx,F}$. Within that working model, maximizing $S_{xx,S}$ minimizes this variance term, which is why the $S_{xx}$ rule is an interpretable baseline rather than an arbitrary heuristic.

Equation (2) is algebraic. Equation (3) is a model-based benchmark, not a coverage guarantee for the observed archive. The fitted residual $e$ can contain real interannual variability, curvature, changing observation support and measurement error; it is not all noise. The benchmark also conditions on the fixed year set and does not represent selection that depends on outcomes or on a different error structure. Table S68 compares the observed year-set RMSE with this benchmark. The two are of similar order in both gap families, so the magnitude of the observed effect alone does not identify a unique thermal mechanism or assign a percentage of the error to geometry. The contribution is the measured consequence of screening, selection and recovery in the river archive, not the general least-squares identity.

### S36.7. Point-estimate accuracy and availability are different questions

The paired accuracy table in the main text is restricted to cases where both D2 and the comparator have a finite forward point estimate under all-year retention. In the controlled-

gap family that common set has 738 archives; D2 RMSE is 0.1638 and the classical comparator RMSE is 0.1466 °C per decade, with a paired D2-minus-classical interval of [0.0036, 0.0270]. In the empirical-calendar family the common set has 701 archives; the corresponding values are 0.0618, 0.0513 and [-0.0028, 0.0178]. The controlled-gap interval lies above zero, while the calendar interval includes zero. These are fixed-budget point-estimate comparisons: they do not establish a universal accuracy ranking, and they do not describe probability-interval availability.

Availability has a different denominator. D2 supplies a finite point estimate on 820/820 forward controlled-gap archives and on 820/820 forward empirical-calendar archives. The classical comparator supplies one on 738/820 and 701/820, respectively. On the 82 controlled-gap and 119 calendar archives where the classical comparator is unavailable but D2 is available, D2 RMSE is 0.2437 and 0.0933 °C per decade. A larger availability denominator therefore does not imply better accuracy on the common cases, and it does not certify that the additional values are suitable for release. Missing classical outputs remain unavailable; they are not assigned zero error. These counts are for the all-year point-estimate layer and must not be substituted for the 805/818 retention-rule cohorts or for probability-interval coverage denominators.

## S37. External reference qualification overview

Strict independent-reference accuracy remains unresolved. Table S69 keeps complete or bounded reference years, stations meeting each history rule, quality and clock, independence, and scored cases in separate fields. Nine Greater Wellington sites stayed at the metadata screen because temperature series were not requested, so their complete hot-year count is unknown (Greater Wellington Regional Council, 2016). The saved screens found zero

strict complete years at 24 NEON sites and zero complete July–June thermal years at five Réunion sites; eight new USGS candidates contain 8 complete station-years, but 0 stations reach four years (Marzolf, 2025; Office français de la biodiversité & Bureau de recherches géologiques et minières, 2026; U.S. Geological Survey, n.d.). That USGS count is not an hourly physical reference and does not prove independence. ECCC Red River shares a river with a source station and retains grade and time-basis questions (Environment and Climate Change Canada, 2024; Redoloza et al., 2025). An Australian candidate archive had prior evaluation exposure: 79 of 80 candidate station-years were previously evaluable, including all 10 designated test years (Australian Bureau of Meteorology, 2026). These exclusions do not establish that the observations are erroneous.

Two exploratory comparisons bound what external evidence can show. British Columbia has 4 stations and 16 station-years. Of 372 masks, 288 are controlled gaps and 84 are calendar gaps. Plug-in, D2 and source-role-calibrated stochastic H30 mean absolute errors on the controlled gaps are 0.516, 0.306 and 0.302 °C (Moore & Callahan, 2024; Callahan & Moore, 2025). Three stations share the Fraser basin; daily means may use only 20 hourly observations, their day convention is unresolved, and source quality processing used water–air comparisons (Water Survey of Canada, n.d.). Nova Scotia covers 2 stations, 8 station-years and 144 masks selected after inspecting availability; D2 and stochastic mean absolute errors are 0.311 and 0.313 °C (Government of Nova Scotia, Environment and Climate Change, 2026). Neither comparison is strict independent-climate validation.

The continuous USGS support at Rio Mora is the clearest warning about target definition. Three years have at least one approved unqualified continuous observation in every fixed-standard-time hour. Only 2017 has every quarter-hour slot. The continuous targets for 2012

and 2019 assign unknown quarter-hour values to the conditional range $[-5, 50]\,^{\circ}\mathrm{C}$; they are not fully verified physical years. The 54 hourly-supported masks give D2 MAE 0.180°C against the published daily target. Using the original histories, D2 interval coverage is 53/54 against published daily H30 and 33/54 against that conditional continuous target; qualifier-aware histories yield 48/54 and 28/54. The difference can combine clock, rounding, corrections, daily preprocessing and the bound assumption. It does not identify a single source error or justify a universal physical calibration. Truckee scores 72 controlled-gap masks on one station and four reported years. The reservoir comparison scores 306 exploratory bounded-reference cases at 4 stations and 17 target station-years. Both remain descriptive checks of bounds and clock assumptions, not independent accuracy tests.

**Table S60.** Fixed-input trajectory sensitivity, 20 to 500 nested draws, by seed and region–reconstructor stratum. Med./max. are absolute 95th-percentile changes in °C. Tail count is the number with either endpoint moving by more than 0.05°C. Intervals are uncalibrated; each stratum is one connected group.

| Seed | Stratum | $n$ | Med./max. | Tail count | Width 20/500 | Covered 20/500 |
|---|---|---|---|---|---|---|
| 0 | Switzerland / donor | 18 | 0.0011/0.1770 | 4 | 0.138/0.157 | 15/16 |
| 0 | Bavaria / air | 9 | 0.1140/0.4346 | 6 | 0.677/0.792 | 8/9 |
| 1 | Switzerland / donor | 18 | 0.0033/0.0887 | 2 | 0.151/0.153 | 16/16 |
| 1 | Bavaria / air | 9 | 0.1057/0.3824 | 6 | 0.700/0.821 | 8/9 |
| 2 | Switzerland / donor | 18 | 0.0043/0.1369 | 5 | 0.153/0.153 | 16/16 |
| 2 | Bavaria / air | 9 | 0.0194/0.2570 | 5 | 0.790/0.831 | 9/9 |

**Table S61.** Estimated slope RMSE sensitivity to weighting and selection opportunity. Archive, station and group columns weight each respective unit equally within the 805-archive cohort; the last column uses only the 468 nontrivial-choice archives and equal archive weights. These are exploratory summaries, not equivalence tests.

| Rule | Archive wt. | Station wt. | Group wt. | Choice-only |
|---|---|---|---|---|
| Warm-season | 0.4459 | 0.4436 | 0.3776 | 0.3944 |
| Uniform exact | 0.4492 | 0.4471 | 0.3806 | 0.4008 |
| CDF | 0.4284 | 0.4262 | 0.3572 | 0.3594 |
| Sxx | 0.4023 | 0.4003 | 0.3413 | 0.3031 |
| All years | 0.1697 | 0.1689 | 0.1524 | 0.1711 |

**Table S62.** Same-target annual comparison by region–reconstructor stratum. D2 and historical native20 are fixed original references. Updated rows regenerate the source-role offsets at the stated nested budget; entries are minima–maxima across seeds 0, 1 and 2, not confidence intervals. MAE and mean 90% width are in °C. CRPS uses the same seven-quantile normal-spacing 256-point approximation for every method. Each stratum contains one connected group; these counts do not estimate population coverage or establish a general method ranking.

| Method / paths | Covered | MAE | Width | CRPS256 |
|---|---|---|---|---|
| *Switzerland / donor ridge; 18 target station-years* | | | | |
| Fixed D2 | 18/18 | 0.0569 | 0.7347 | 0.0483 |
| Historical native20 | 18/18 | 0.0785 | 0.5623 | 0.0554 |
| Updated nested20 | 18 of 18 | 0.0632–0.0666 | 0.5802–0.6716 | 0.0504–0.0509 |
| Updated nested100 | 18 of 18 | 0.0642–0.0724 | 0.5677–0.5715 | 0.0490–0.0521 |
| Updated nested500 | 18 of 18 | 0.0693–0.0730 | 0.5334–0.5533 | 0.0500–0.0516 |
| *Bavaria / air ridge; 9 target station-years* | | | | |
| Fixed D2 | 9/9 | 0.2311 | 1.0531 | 0.1634 |
| Historical native20 | 9/9 | 0.2435 | 0.7612 | 0.1530 |
| Updated nested20 | 8–9 of 9 | 0.1914–0.2321 | 0.7357–0.8804 | 0.1281–0.1621 |
| Updated nested100 | 8–9 of 9 | 0.1824–0.2129 | 0.7526–0.8547 | 0.1262–0.1467 |
| Updated nested500 | 9 of 9 | 0.1915–0.2015 | 0.7607–0.8006 | 0.1287–0.1334 |

**Table S63.** Updated source-offset annual results for every nested seed and budget. Panels retain the original region–reconstructor strata. Counts are descriptive on one group per panel; all scores use the same bounded 256-point annual approximation.

| Seed / paths | $n$ | Covered | MAE | Width | CRPS256 |
|---|---|---|---|---|---|
| *Switzerland / donor ridge* | | | | | |
| 0 / 20 | 18 | 18 | 0.0666 | 0.5802 | 0.0504 |
| 0 / 100 | 18 | 18 | 0.0700 | 0.5677 | 0.0521 |
| 0 / 500 | 18 | 18 | 0.0730 | 0.5334 | 0.0513 |
| 1 / 20 | 18 | 18 | 0.0649 | 0.6716 | 0.0507 |
| 1 / 100 | 18 | 18 | 0.0724 | 0.5715 | 0.0513 |
| 1 / 500 | 18 | 18 | 0.0700 | 0.5533 | 0.0516 |
| 2 / 20 | 18 | 18 | 0.0632 | 0.6092 | 0.0509 |
| 2 / 100 | 18 | 18 | 0.0642 | 0.5711 | 0.0490 |
| 2 / 500 | 18 | 18 | 0.0693 | 0.5503 | 0.0500 |
| *Bavaria / air ridge* | | | | | |
| 0 / 20 | 9 | 8 | 0.2321 | 0.7949 | 0.1621 |
| 0 / 100 | 9 | 8 | 0.2129 | 0.7526 | 0.1467 |
| 0 / 500 | 9 | 9 | 0.2015 | 0.7607 | 0.1334 |
| 1 / 20 | 9 | 9 | 0.1914 | 0.7357 | 0.1281 |
| 1 / 100 | 9 | 9 | 0.1845 | 0.7643 | 0.1262 |
| 1 / 500 | 9 | 9 | 0.1925 | 0.7740 | 0.1323 |
| 2 / 20 | 9 | 9 | 0.1960 | 0.8804 | 0.1308 |
| 2 / 100 | 9 | 9 | 0.1824 | 0.8547 | 0.1317 |
| 2 / 500 | 9 | 9 | 0.1915 | 0.8006 | 0.1287 |

**Table S64.** Both endpoint sensitivities after source-role offset re-estimation, from nested-prefix20 to 500 paths. Entries are the median/maximum absolute change in °C across the stated target stratum; the last column counts cases with either endpoint changing by more than 0.05°C. Each region–reconstructor stratum is one connected group, and these descriptive changes do not establish convergence.

| Stratum | Seed | $n$ | $\lvert\Delta q_{.05}\rvert$ med./max. | $\lvert\Delta q_{.95}\rvert$ med./max. | Either $> 0.05$ |
|---|---|---|---|---|---|
| Swiss donor | 0 | 18 | 0.0000/0.1353 | 0.0691/0.1159 | 11/18 |
| Swiss donor | 1 | 18 | 0.0000/0.0561 | 0.1169/0.2619 | 13/18 |
| Swiss donor | 2 | 18 | 0.0000/0.0388 | 0.1024/0.2667 | 12/18 |
| Bavarian air | 0 | 9 | 0.0099/0.2223 | 0.0609/0.3688 | 7/9 |
| Bavarian air | 1 | 9 | 0.0433/0.2918 | 0.1011/0.3140 | 6/9 |
| Bavarian air | 2 | 9 | 0.0034/0.4496 | 0.0670/0.2367 | 7/9 |

**Table S65.** Influence checks. Panel A shows FOEN-minus-Bavaria coverage gaps in percentage points using min-count weights; group deletion recomputes common support. Panel B shows after-minus-before-2010 within-station annual slopes; equal-station weighting changes the target relative to equal annual observations. Deletion ranges are not confidence intervals.

| Panel A: cells | Main gap | Group-deletion gap | Sign reversals |
|---|---|---|---|
| Geometry | 16.037 | 12.392–17.772 | 0 |
| + reconstructor | 3.686 | -1.120–5.292 | 2 |
| **Panel B: region** | **Annual-weight Δ** | **Station-weight Δ** | **Group-deletion Δ** |
| Switzerland | -0.0143 | -0.0137 | -0.0152 to -0.0047 |
| Bavaria | -0.0054 | -0.0343 | -0.0882–0.1089 |

**Table S66.** Same-day overlap denominators. Any overlap is the percent of peer-evaluable incomplete rows with at least one shared missing day. Day share uses eligible missing station-days. Repeated trial masks retain their original weights; the historical observed row supplements the primary 2010–2025 comparison.

| Source / period | All | Peer | Excl. | Any (%) | Shared / eligible days | Days (%) |
|---|---|---|---|---|---|---|
| Observed / 2010–25 | 109 | 74 | 35 | 18.92 | 40/2,086 | 1.92 |
| Calendar / 2010–25 | 3,806 | 2,810 | 996 | 36.01 | 4,946/41,507 | 11.92 |
| Controlled / 2010–25 | 3,777 | 2,779 | 998 | 58.58 | 64,164/165,870 | 38.68 |
| Observed / 1981–25 | 220 | 147 | 73 | 32.65 | 158/3,946 | 4.00 |

**Table S67.** Post-review calendar-position extension on saved annual predictions. Control-gap and empirical-calendar families are reported separately; their rows are not a common cohort and their differences must not be subtracted to name a missingness mechanism. Warm-season is the original 90% warm-season rule; the legacy one-draw random and incomplete-year-only spacing controls are not repeated here. Uniform exact averages squared errors over every feasible same-count set within each archive, then takes the cohort root. CDF and $S_{xx}$ use calendar years only and keep every fixed complete year. Intervals resample the 17 connected groups 10,000 times; they are conditional on the saved predictions and the rules compared.

| Gap family / rule | Archives | Est. RMSE | Year-set RMSE | Reconst. RMSE | Δ RMSE [95% interval] |
|---|---|---|---|---|---|
| *Controlled 30–90 day; 805 actual-mask archives* | | | | | |
| Warm-season | 805 | 0.4459 | 0.4458 | 0.0023 | Reference |
| Uniform exact | 805 | 0.4492 | 0.4483 | 0.0978 | 0.0033 [-0.0051, 0.0172] |
| CDF | 805 | 0.4284 | 0.4199 | 0.1079 | -0.0175 [-0.0274, -0.0052] |
| Sxx | 805 | 0.4023 | 0.3931 | 0.1185 | -0.0436 [-0.0649, -0.0172] |
| All years | 805 | 0.1697 | 0.0000 | 0.1697 | -0.2762 [-0.3574, -0.1839] |
| *Empirical calendar; 818 actual-mask archives* | | | | | |
| Warm-season | 818 | 0.1136 | 0.1120 | 0.0150 | Reference |
| Uniform exact | 818 | 0.1355 | 0.1223 | 0.0695 | 0.0219 [-0.0036, 0.0369] |
| CDF | 818 | 0.0984 | 0.0727 | 0.0684 | -0.0152 [-0.0451, 0.0004] |
| Sxx | 818 | 0.0859 | 0.0558 | 0.0679 | -0.0277 [-0.0497, -0.0169] |
| All years | 818 | 0.0674 | 0.0000 | 0.0674 | -0.0462 [-0.0706, -0.0295] |

**Table S68.** Observed year-set RMSE and the iid linear-reference working benchmark, in °C per decade. The benchmark uses each archive's complete-reference residual variance and the exact weight-distance formula. It is an assumption-dependent scale check, not a fitted hydrological model, a coverage guarantee or a decomposition of the observed error.

| Gap family | Rule | Observed year-set RMSE | iid working benchmark |
|---|---|---|---|
| Controlled 30–90 day | Warm-season | 0.4458 | 0.4865 |
| Controlled 30–90 day | $S_{xx}$ | 0.3931 | 0.4193 |
| Empirical calendar | Warm-season | 0.1120 | 0.1616 |
| Empirical calendar | $S_{xx}$ | 0.0558 | 0.0639 |

**Table S69.** External reference qualification. Complete or bounded reference years, stations meeting the stated history rule, quality and clock, independence, and scored cases are separate fields. Unknown is written as unknown. A zero appears only when the saved screen counted that quantity.

| Source | Reference years | History-rule stations | Quality and clock | Independence | Scored cases |
|---|---|---|---|---|---|
| Greater Wellington | Temperature series not acquired | 9 potential sites; quality-code screen not completed | Quality-code values were not requested | Unknown | Not scored |
| NEON | 0 strict complete years | 0 of 24 sites meet the four-year strict rule | UTC day and final quality flag required | Climate independence not established | Not scored |
| New USGS candidates | 8 complete station-years | 0 of 8 sites reach four complete years | Daily screen only; hourly reference not verified here | Not established | Not scored. The observation-only witness check is separate. |
| Réunion (Hub'Eau) | 0 complete July–June thermal years | 0 of 5 sites meet the four-year rule | Reported-day time zone unresolved | Unknown | Not scored |
| ECCC Red River | 0 complete local wall-clock years | Strict physical reference not met | Grade and time basis unresolved | Shares a river with a source station | Not scored |
| Australia candidate | 80 candidate station-years; strict 24-hour sampling not established | 79 of 80 were previously evaluable, including 10 of 10 designated test years | The saved screen allowed 20 observed hours | Not independent: prior evaluation exposure | Not scored as a new test |
| British Columbia | 16 published station-years; 0 strict 24-hour stations | 4 stations scored; 0 meet the strict 24-hour rule | Days may contain 20 hours; clock unresolved | Not established; three stations share the Fraser basin | 372 masks total: 288 controlled gaps and 84 calendar gaps. The reported MAEs use the 288. |
| Nova Scotia | 8 reported-clock station-years | 2 stations; strict reference not met | Source QC and time zone unresolved | Not established | 144 controlled-gap masks scored |
| Truckee, California | 4 reported years | 1 station; strict reference not met | Clock and estimation flags unresolved | Not established; one managed station with prior basin contact | 72 controlled-gap masks scored |
| USGS Rio Mora | Complete quarter-hour year: 2017. Bounded years: 2012 and 2019 (2012: 3 unknown slots; 2017: 0 unknown slots; 2019: 2 unknown slots) | 1 station | Unknown quarter-hour slots are assigned to the conditional range $[-5, 50]\,^{\circ}\mathrm{C}$ | Not an independent climate | 54 hourly-supported masks scored against the reported daily target and the conditional continuous target |
| Reservoir network | 0 complete finite years among 69 station-years | 0 of 10 files have four finite years | Strict quarter-hour calendars were not met. The scored bounds assume missing values lie in $[-5, 50]\,^{\circ}\mathrm{C}$ | Not established | 306 exploratory bounded-reference cases scored at 4 stations and 17 target station-years |

## S38. Post-hoc retained-truth and point-summary checks

These checks reuse the saved forward and offline predictions. They do not refit models, open new stations, or supply an independent validation. Replacing a retained annual value by the experimental truth cannot be done when that year is actually missing.

On each common station–repetition, the warm-season year set is held fixed and every retained D2 annual value is replaced by the known complete-record H30 before the slope is recomputed. The resulting root-mean-square error is the error of the year set alone. Table S70 shows that this truth-in-set error is nearly the same as the error of the saved D2 values, especially for forward controlled gaps (0.445242 versus 0.445331 °C per decade). For a finite complete record, the warm-season screen changes the slope mainly by changing which years and ordinary-least-squares weights remain. Making the retained years exact does not restore the years that were removed.

The two root-mean-square errors are not subtracted, and neither is divided by the other to name a percentage caused by retention. The squared error of the warm-season slope relative to the complete-record slope also contains the cross term between the year-set change and the reconstruction error on the retained years. The four-cell symmetric decomposition in the main text keeps its original definition. The target remains the slope of a particular complete archive, not a latent climate-process trend.

The main analysis publishes the annual median and then scores the trend of that published series. A conditional mean is the relevant point for squared error of a slope when the conditional distribution is correct; the median does not automatically have that property. Table S71 replaces only the annual point, using the mean of the same approximate distribution on the 738 archives where both methods have a forward controlled-

gap result. The mean still depends on interpolation between the seven quantiles and on the normal-spacing tails. Both root-mean-square errors change slightly, and the original ordering does not reverse. This is not a claim that the mean is a verified optimum, and it does not replace the published-median analysis.

**Table S70.** D2 slope error when retained years keep their saved annual values or are replaced by the known complete-record H30. Units are °C per decade. The truth replacement is an evaluation device.

| Gaps | Timing | Common settings | All years | Warm season | Warm season, true H30 |
|---|---|---|---|---|---|
| Calendar | Offline | 840 | 0.008732 | 0.024041 | 0.023985 |
| Controlled | Offline | 840 | 0.027772 | 0.100603 | 0.100603 |
| Calendar | Forward | 820 | 0.067277 | 0.113458 | 0.111852 |
| Controlled | Forward | 807 | 0.169474 | 0.445331 | 0.445242 |

**Table S71.** Trend root-mean-square error, in °C per decade, when the annual point is the published median or the mean of the same approximate distribution. Both use the 738 common forward controlled-gap archives.

| Method | Published median | Approximate-distribution mean |
|---|---|---|
| D2 | 0.163814 | 0.157811 |
| Calibrated stochastic | 0.146577 | 0.143719 |

## S39. Independent observed-record arithmetic re-audits

Using the same local canonical float32 daily archive and an independent convolution implementation, we recomputed 2,139 complete reference station-years and 545 incomplete station-years at 96 eligible stations. All 35 witness identities, saved 30-day values and complete-year reference maxima agreed within the stated numerical tolerance. This is an arithmetic re-audit of the same processed archive, not an independent check of provider-native records or sensor error.

Comparing 971,673 finite values from 34 selected project input panels with the canonical archive yielded no missingness or float32-cast discrepancies. The largest absolute rounding difference is 0.00000153 degrees C; 0 witness identities and 0 classifications at the 0.5-degree excess threshold changed. The input panels are processed project files, not provider-native instrument records.

Across 2,700 eligible station-years, independently recalculated annual coverage, available 30-day maxima, July–August completeness and longest missing runs reproduced 210 witness–rule decisions. For each of 35 witness years, increasing its observed values by 5 degrees C did not change the leave-target-year-out warm-season climatology. The old fixed-season fraction remains sourced from saved metadata; these probes do not cover every year or other pipelines.

For the ten additional USGS sites, independent parsing of the saved daily API responses reproduced 8 approved-complete reference station-years and the 9 certified observed windows. The original quality-screened window selection also replays from the saved payloads, but the separate arithmetic audit does not independently implement that selection. Approved estimated values remain in complete-year references, but the certified

windows contain no estimated days. These records do not establish subdaily completeness, physical measurement accuracy or independent recovery performance.

A separate candidate review subset contains processed daily values for 95 of 96 previously eligible source stations, including all 35 witnesses. One LUBW station is omitted pending permission. The subset verifies the included reported-series arithmetic and rule decisions, but cannot independently confirm the omitted station or the original full eligible-station denominator.

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